

Design and Develop a Centralized Freelancing Platform

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of

Bachelor of Technology

in

Computer Science and Engineering

by

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BUDGE BUDGE INSTITUTE OF TECHNOLOGY

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CERTIFICATE OF APPROVAL

This is to certify that the major project entitled “**Design and Develop a Centralized Freelancing Platform**” submitted by **Bijoy Pantu, Sharad Kumar, Rishika Kumari, Koustav Mondal, Muntazir Alam**, to the Department of Computer Science and Engineering, Budge Budge Institute of Technology, Kolkata, is a record of Bonafide research work carried out under my supervision and is worthy of consideration for the award of B.Tech. degree from Maulana Abul Kalam Azad University of Technology, formerly WBUT.

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We hereby declare that the project work entitled “**Design and Develop a Centralized Freelancing Platform**” submitted to **Budge Budge Institute of Technology** is our original work, carried out under the guidance and supervision of **Mrs. Atreyee Datta** in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology (B.Tech.) in Computer Science & Engineering.

We further declare that this project report has not been submitted, either in whole or in part, to any other university or institution for the award of any degree, diploma, or other academic qualification. All sources of information and references used in this report have been duly acknowledged.

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To the best of my knowledge, this report has not been submitted for any other award. The student’s performance during the project period has been satisfactory.

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With regards

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ABSTRACT

The rapid expansion of the digital economy and the increasing adoption of remote work have significantly boosted the growth of freelancing and gig-based employment, especially in a developing country like India. Despite the availability of global freelancing platforms, there exists a noticeable gap in providing localized, secure, and intelligent platforms that effectively address the needs of Indian freelancers and employers. Existing international platforms often come with high commission fees, currency conversion issues, and payment delays, which discourage many local freelancers and small businesses from using them effectively. This project focuses on the design and development of a **Freelancing Platform** aimed at bridging this gap.

The proposed system serves as an online marketplace that connects freelancers with employers seeking short-term, gig-based, or project-oriented services. The platform enables freelancers to create detailed profiles highlighting their skills, experience, and portfolio, while employers can post projects, define requirements, and select suitable candidates. Advanced features such as AI-based job recommendations, intelligent search and analytics, rating and review mechanisms, and escrow-based payment systems are integrated to ensure trust, transparency, and efficiency. The system also emphasizes ease of use through a clean interface, ensuring that users with varying levels of technical expertise can navigate the platform without difficulty.

The outcome of this project is a scalable, user-friendly, and secure freelancing platform that enhances employment opportunities, improves job satisfaction, and contributes to the growth of India's gig economy by offering a dependable, locally relevant alternative to existing global solutions.

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CHAPTER 1: INTRODUCTION

1.1 Introduction to Freelancing Platform

A freelancing platform is a digital marketplace that enables independent professionals, commonly referred to as freelancers, to offer their skills and services to businesses, organizations, and individual clients on a temporary, project-based, or contractual basis, rather than through conventional full-time employment. Unlike traditional employment structures, where an individual is tied to a single employer under a fixed salary and schedule, freelancing allows professionals the flexibility to choose their own projects, negotiate their own rates, manage their own working hours, and often work remotely from any location with an internet connection. This shift in the nature of work has been driven by rapid advancements in digital communication technologies, cloud-based collaboration tools, and secure online payment systems, all of which have made it increasingly feasible for freelancers and employers to interact, negotiate, and transact without the need for physical proximity.

Over the past decade, the concept of freelancing has evolved from being a peripheral or supplementary source of income to becoming a mainstream and, in many cases, primary career choice for millions of individuals across the globe. This evolution has been particularly significant in the aftermath of the COVID-19 pandemic, which acted as a catalyst in normalizing remote work culture and demonstrating that a large proportion of professional tasks, particularly in fields such as software development, graphic design, content writing, digital marketing, and data analysis, could be performed effectively outside the boundaries of a conventional office environment. As a result, organizations of all sizes, from small startups to large multinational corporations, have increasingly turned to freelance talent to meet short-term project requirements, reduce overhead costs associated with full-time hiring, and gain access to specialized skills that may not be readily available within their existing workforce.

India, in particular, has emerged as one of the largest and fastest-growing freelance markets in the world. With a vast population of technically skilled graduates, a growing number of engineering and computer science professionals, and increasing internet and smartphone penetration even in semi-urban and rural areas, the country has witnessed a substantial rise in the number of individuals choosing freelancing as either a primary occupation or a secondary source of income alongside their regular jobs. Reports on the global gig economy consistently place India among the top contributing nations in terms of the number of registered freelancers on international platforms. This growing workforce spans a wide range of domains, including web and mobile application development, UI/UX design, content creation, digital marketing, virtual assistance, video editing, and data entry, among many others.

Despite this significant participation, a large percentage of Indian freelancers continue to rely on international freelancing platforms such as Upwork, Fiverr, and Freelancer.com to find work opportunities. While these platforms have undoubtedly played an important role in connecting freelancers with global clients, they are often not optimally designed to address the specific needs, expectations, and challenges faced by users operating within the Indian context. Common issues include high service and withdrawal fees that eat into a freelancer's earnings, complications arising from currency conversion when payments are made in foreign currencies, delayed fund transfers due to international banking procedures, and a lack of localized job categories that reflect the specific skill sets and service demands prevalent in the Indian market. Additionally, new freelancers often face intense competition from a global pool of highly experienced professionals, making it difficult for them to establish a foothold and build a reliable client base, particularly in the early stages of their freelancing careers.

This project proposes the design and development of a comprehensive, centralized freelancing platform conceptually similar to established global platforms like Upwork, but specifically tailored to better serve the Indian freelancing ecosystem. The platform is envisioned as a secure, intelligent, and user-friendly online marketplace that facilitates seamless interaction between freelancers and employers. Freelancers using the platform will be able to create detailed professional profiles that showcase their skills, work experience, educational qualifications, and portfolios of previous work, thereby enhancing their credibility and visibility to potential clients. Employers, on the other hand, will be provided with the tools necessary to post detailed project requirements, specify budgets and timelines, review freelancer profiles and past work samples, and ultimately select the most suitable candidates for their specific needs.

By addressing the specific pain points associated with existing international platforms and introducing features tailored to the Indian context, this project aims to contribute meaningfully to the growth of the domestic gig economy. It seeks to empower a new generation of Indian freelancers by providing them with a reliable, transparent, and cost-effective avenue to showcase their talents and secure meaningful work opportunities, while simultaneously offering employers a trustworthy platform through which they can access a rich and diverse pool of skilled professionals to meet their project-based staffing needs.

1.2 Literature Review

The global freelancing industry is currently dominated by a handful of well-established platforms, including Upwork, Fiverr, Freelancer.com, and Toptal, each of which has adopted a distinct approach to connecting freelancers with clients. Upwork operates primarily on a bidding and proposal-based model, allowing freelancers to apply directly to job postings with customized proposals, and supports both hourly and fixed-price contracts along with built-in time-tracking tools. Fiverr, in contrast, follows a service-catalogue

model, where freelancers create predefined "gigs" at fixed price tiers that clients can browse and purchase directly, making it particularly popular for smaller, well-defined tasks. Freelancer.com combines elements of both bidding and contests, allowing clients to award projects based on competitive proposals or through skill-based contests, while Toptal positions itself as an exclusive, high-end platform that screens and admits only a small percentage of applicants, catering primarily to enterprise clients seeking top-tier talent.

While these platforms have collectively created a robust global infrastructure for remote work and have enabled millions of freelancers to find international clients, a review of user feedback, industry reports, and academic literature on the gig economy reveals several recurring limitations, particularly from the perspective of freelancers based in India and other developing economies. Commonly cited concerns include high service and withdrawal fees, which can range from ten to twenty percent of a freelancer's earnings, payment delays caused by cross-border banking procedures and currency conversion, and an intensely competitive environment where new freelancers struggle to secure their first few projects due to the sheer volume of established, highly-rated competitors on these platforms. Additionally, existing platforms rarely offer job categories or search filters tailored to skill sets and pricing norms specific to the Indian market, and their dispute resolution and customer support systems are often not well-suited to address region-specific concerns.

Studying these existing systems has provided valuable insight into both the strengths worth retaining, such as structured profile-building, rating and review systems, and secure payment handling, and the gaps that present an opportunity for a more localized solution. In particular, this review highlighted a clear need for a platform that incorporates local and low-cost payment gateways, an escrow-based payment protection mechanism to build trust between new users.

1.3 Literary Survey / Existing systems survey

Upwork

Description: A globally recognized freelancing platform that connects businesses with freelancers across categories like web development, design, writing, and marketing. Supports both hourly and fixed-price contracts with built-in time tracking and invoicing.

Limitations: Charges high service fees (up to 20%) on freelancer earnings. Intense competition makes it difficult for new freelancers to get their first project. Lacks India-specific payment gateways.

Official Site: <https://www.upwork.com/>

Fiverr

Description: A service-catalogue based platform where freelancers list predefined "gigs" at fixed price tiers that clients can browse and purchase directly. Popular for smaller, well-defined tasks and creative services.

Limitations: Limited scope for negotiation on larger or custom projects. Payment withdrawal can take several days to clear. Low starting gig prices lead to undervaluation of freelancer skills.

Official Site: <https://www.fiverr.com/>

Freelancer.com

Description: Combines bidding-based project proposals with skill-based contests, allowing clients to award work either through competitive bids or contest submissions.

Limitations: High number of low-quality bids can bury genuine proposals. Membership tiers restrict the number of bids for free users. Customer support is often slow to resolve disputes.

Official Site: <https://www.freelancer.com/>

Toptal

Description: An exclusive, high-end freelancing platform that screens and admits only a small percentage of applicants, primarily catering to enterprise clients seeking top-tier, vetted talent.

Limitations: Extremely selective admission process excludes most freelancers, especially beginners. Primarily suited for large-budget clients, not small businesses or individuals. No open marketplace browsing.

Official Site: <https://www.toptal.com/>

1.4 Problem Statement

Although freelancing has emerged as one of the fastest-growing employment models in India, there remains a lack of a dedicated, centralized platform that efficiently addresses the specific needs of Indian freelancers and employers. Existing global platforms, while widely used, are not designed with the Indian market in mind, resulting in several persistent challenges. Freelancers frequently encounter high service fees that reduce their net earnings, delayed and complicated payment withdrawals due to cross-border banking processes, and an overwhelming level of competition from a global pool of established professionals, making it especially difficult for newcomers to secure their first few projects. At the same time, employers looking to hire freelancers for short-term or project-based work often struggle to find candidates whose skills, pricing expectations, and availability align well with region-specific requirements and budgets.

Beyond these financial and competitive barriers, existing platforms generally lack robust mechanisms to ensure trust and transparency between freelancers and employers, particularly for users who are new to the platform and have not yet built a verifiable track record. The absence of secure, escrow-based payment protection on many platforms leaves both parties vulnerable to fraud, non-payment, or unsatisfactory work delivery, further discouraging participation from first-time users. Additionally, the job discovery process on many existing platforms relies on basic keyword-based search and manual browsing, rather than intelligent, skill-based matching, which often results in freelancers spending excessive time and effort searching for suitable opportunities, while employers struggle to identify the most qualified candidates from a large and often irrelevant pool of applicants.

These challenges collectively highlight a clear and pressing need for a dedicated freelancing platform tailored to the Indian ecosystem, one that offers localized payment solutions, transparent and secure transaction handling, and intelligent job-matching capabilities. This project aims to directly address these gaps by developing a robust, secure, and user-friendly freelancing platform that ensures fair compensation, builds trust between users, and simplifies the process of connecting the right freelancers with the right employers.

1.5 Objectives of the Project

The primary aim of this project is to design and develop a centralized, secure, and intelligent freelancing platform that effectively bridges the gap between Indian freelancers and employers seeking short-term, gig-based, or project-oriented services. To achieve this overarching goal, the project has been structured around a set of clearly defined primary and secondary objectives, each of which addresses a specific aspect of the platform's functionality, usability, and long-term impact on its intended user base.

1.5.1 Primary Objectives

The primary objectives represent the core, non-negotiable goals that the platform must fulfil in order to be considered a viable and functional freelancing solution. These objectives form the foundation upon which all other features and enhancements are built.

- To design and develop a fully functional, web-based freelancing platform that facilitates project-based and gig employment by connecting freelancers with employers in a structured and organized manner.
- To ensure secure and reliable interaction between freelancers and employers throughout the entire engagement lifecycle, from initial project discovery and communication to final payment settlement, thereby minimizing the risk of fraud, miscommunication, or disputes.

- To establish a transparent and fair payment mechanism that protects the financial interests of both freelancers and employers, ensuring that freelancers are compensated promptly for completed work and employers receive the agreed-upon deliverables before funds are released.
- To create an intuitive and accessible user interface that allows individuals with varying levels of technical proficiency to navigate the platform, create profiles, post or apply for projects, and manage ongoing engagements without unnecessary complexity.
- To build a scalable system architecture capable of accommodating a growing number of users, projects, and transactions over time, ensuring that the platform remains performant and reliable as its user base expands.

1.5.2 Secondary Objectives

In addition to the primary objectives, several secondary objectives have been identified to enhance the overall functionality, competitiveness, and user experience of the platform. While not strictly essential to its basic operation, these objectives significantly improve the platform's value proposition and differentiate it from existing freelancing solutions.

- To enable freelancers to build detailed, comprehensive profiles that showcase their skills, work experience, educational background, certifications, and portfolios of previously completed projects, thereby improving their visibility and credibility to prospective employers.
- To integrate an AI-based recommendation system capable of intelligently matching freelancers with relevant job postings based on their skill sets, past performance, and stated preferences, reducing the time and effort required for job discovery on both sides of the platform.
- To implement advanced search and filtering functionality, supported by analytics dashboards, allowing both freelancers and employers to efficiently locate relevant opportunities or candidates based on specific criteria such as skill category, budget range, project duration, and location.
- To implement a secure, escrow-based payment mechanism that holds client funds in trust until project milestones are verified and approved, thereby providing financial protection and building confidence among first-time users of the platform.
- To incorporate a comprehensive rating and review system that allows freelancers and employers to evaluate one another following the completion of a project, fostering accountability, encouraging quality work, and helping new users build a trustworthy reputation over time.

- To design the platform with future scalability in mind, allowing for the potential integration of additional features such as real-time chat, video-call-based interviews, dispute resolution mechanisms, and mobile application support in subsequent development phases.

Collectively, these primary and secondary objectives guide the overall design, development, and implementation strategy adopted throughout this project, ensuring that the resulting platform is not only functional and secure, but also competitive, user-centric, and well-positioned to serve the evolving needs of the Indian freelancing ecosystem.

1.6 Features of the Project

The freelancing platform is designed around a core set of features that together form a complete ecosystem supporting the entire freelance engagement lifecycle, from project discovery to final payment.

- **Freelance job marketplace** - a centralized space where employers post projects across various categories and freelancers discover relevant opportunities suited to their skills.
- **Employer project posting and freelancer bidding** - employers can define project scope, budget, and timeline, while freelancers submit competitive proposals to secure work.
- **Freelancer profile and portfolio management** - freelancers can build detailed profiles showcasing their skills, work experience, certifications, and samples of past projects, improving their credibility to potential employers.
- **AI-powered job recommendations** - an intelligent recommendation engine matches freelancers with the most relevant job postings based on their skill sets, experience, and historical performance, reducing time spent searching for suitable work.
- **Advanced search and analytics** - robust search and filtering tools, backed by analytics dashboards, allow both freelancers and employers to efficiently locate opportunities or candidates based on specific criteria such as budget, category, or duration.
- **Secure escrow account and payment gateway integration** - client funds are held securely in escrow and released only once project milestones are verified, protecting both freelancers and employers from payment disputes or fraud.
- **Rating and review system** - a two-way feedback mechanism allows freelancers and employers to rate one another after project completion, encouraging accountability and helping new users establish a trustworthy reputation over time.

1.7 Scope of the Application

The scope of this freelancing platform extends across a wide range of industries and professional domains, making it a versatile solution rather than one limited to a single niche. It is designed to cater to sectors such as software development, web and mobile application design, graphic design, digital marketing, content writing, video editing, data analysis, virtual assistance, and consulting, among many others. This broad applicability ensures that the platform can serve a diverse user base, ranging from independent freelancers offering specialized technical services to small businesses and startups seeking cost-effective, project-based talent for short-term needs.

The platform has also been designed with long-term scalability in mind. While the initial implementation focuses on establishing core functionalities such as job posting, freelancer bidding, secure payments, and profile management, the underlying architecture is intended to support future expansion into more specialized or niche freelancing verticals, such as legal consulting, academic tutoring, or creative arts, without requiring a fundamental redesign of the system. Similarly, the platform can be scaled from serving a small, regional user base to supporting a much larger, potentially nationwide or even global freelance ecosystem, accommodating millions of freelancers and employers as adoption grows.

Furthermore, the scope of the project extends beyond the core web-based marketplace to include the potential integration of complementary features in future iterations, such as mobile application support for on-the-go access, real-time messaging and video-call functionality for client-freelancer communication, and dispute resolution mechanisms to handle disagreements over deliverables or payments. This forward-looking design philosophy ensures that the platform is not merely a static solution to the problems identified in this project, but a foundation capable of evolving alongside the changing needs of the Indian freelancing landscape.

CHAPTER 2: ANALYSIS AND DESIGN

2.1 Background Research

Before proceeding with the design and development of the platform, extensive background research was conducted to understand the current landscape of the freelancing industry, both globally and within the Indian context. This research involved studying prevailing freelancing trends, the growing market demand for gig-based employment, common employer expectations when hiring freelance talent, and the recurring challenges faced by freelancers in securing consistent, fairly compensated work.

Data for this research was gathered through multiple sources, including online surveys directed at freelancers and small business owners, published industry reports on the gig economy in India, and a detailed analysis of existing freelancing platforms such as Upwork, Fiverr, and Freelancer.com, as covered in the literature review. In addition, informal discussions with individuals who had prior freelancing experience provided practical, first-hand insight into the day-to-day difficulties encountered when using existing platforms, particularly around payment delays, trust issues with new clients, and the difficulty of standing out in a saturated marketplace.

The research consistently highlighted three critical areas that any successful freelancing platform must address: trust, payment security, and intelligent job matching. Trust emerged as a foundational concern, since both freelancers and employers are often hesitant to engage with unfamiliar parties without some assurance of reliability and accountability. Payment security was identified as equally important, with users expressing strong preference for mechanisms that guarantee fair and timely compensation while protecting against fraud or non-delivery. Finally, intelligent job matching was recognized as a key differentiator, as it directly reduces the time and effort required for freelancers to find suitable opportunities and for employers to identify qualified candidates.

These findings directly shaped the core design philosophy of the proposed platform, reinforcing the decision to incorporate features such as escrow-based payments, rating and review systems, and AI-driven recommendations as central pillars of the system rather than optional add-ons.

2.2 Requirement Analysis

The requirement analysis phase forms a critical stage in the development of the freelancing platform, as it establishes a clear understanding of what the system must accomplish and how it should behave under various conditions. This phase involved a systematic identification of both functional and non-functional

requirements, derived from the background research, literature review, and an understanding of the day-to-day needs of freelancers and employers.

The functional requirements define the specific actions and capabilities the system must support to fulfil its core purpose. These include user registration and authentication for both freelancers and employers, the ability for employers to post detailed project listings with defined budgets and timelines, a bidding and proposal submission mechanism for freelancers, tools for managing ongoing projects and tracking milestones, and a secure payment processing system that supports escrow-based transactions. Additionally, functional requirements extend to profile and portfolio management for freelancers, an AI-based job recommendation engine, advanced search and filtering capabilities, and a rating and review system for post-project feedback.

The non-functional requirements, on the other hand, describe the quality attributes and constraints under which the system must operate. Scalability was identified as essential, ensuring the platform can accommodate a growing number of users and transactions without performance degradation. Security requirements demand robust protection of sensitive user data and financial transactions through encryption and secure authentication protocols. Reliability requirements ensure consistent system uptime and accurate data handling, while performance requirements dictate that core operations, such as search queries and payment processing, must execute with minimal latency to maintain a smooth user experience.

2.3 System Requirements

The system requirements define both the functional capabilities and the underlying technical specifications necessary for the freelancing platform to operate reliably as a complete, real-time marketplace connecting freelancers with employers. These requirements span software, hardware, and infrastructure considerations that were evaluated to ensure the platform performs efficiently under realistic usage conditions.

Functional System Components

- **User authentication and authorization:** A secure registration and login mechanism distinguishing freelancer and employer accounts, supported by password hashing (e.g., bcrypt), JSON Web Token (JWT) based session management, and role-based access control.
- **Freelancer and employer dashboards:** Role-specific dashboards providing a centralized, real-time view of active applications, ongoing projects, earnings, and hiring activity.
- **Job posting and application management:** Tools for creating detailed project listings and submitting proposals, with status tracking and in-platform communication between parties.

- **Secure escrow and payment gateway:** Integration with a payment processor (such as Razorpay or Stripe) supporting escrow-style fund holding, with funds released only upon milestone verification.

Software Requirements

- **Frontend:** Built using React.js with Tailwind CSS for a modern, responsive, single-page application experience, compatible with all major browsers including Chrome, Firefox, Edge, and Safari.
- **Backend:** A RESTful API layer built using Node.js with the Express.js framework, handling business logic, authentication, and communication with the database and third-party services.
- **Database:** MongoDB, a NoSQL document-oriented database, used to store user profiles, project listings, proposals, and transaction records in a flexible, schema-less format well-suited to the varied and evolving data structures of a freelancing platform.
- **Operating System Compatibility:** The platform, being web-based, must be accessible across Windows, macOS, and Linux systems through any standard modern web browser, with no dependency on a specific operating system.

Hardware and Server Requirements

- **Development environment:** A system with a minimum of 8GB RAM, a multi-core processor, and at least 20GB of free storage is recommended for smooth local development and testing.
- **Server/hosting infrastructure:** Cloud-based hosting (such as AWS, Azure, or Render) with auto-scaling capability to handle variable traffic loads, minimum 2 vCPUs and 4GB RAM for the initial deployment, with provisions to scale vertically or horizontally as user load increases, alongside a MongoDB Atlas instance or equivalent managed database service.
- **Network requirements:** A stable internet connection with sufficient bandwidth to support real-time features such as notifications, messaging, and file uploads (e.g., portfolio documents, project deliverables) without noticeable latency.

2.4 User Requirements

While the system requirements outlined the technical and functional backbone of the platform, the user requirements focus specifically on what each category of user, namely employers and freelancers, needs to be able to accomplish through the platform in order to meet their respective goals effectively. These requirements were derived from studying the typical workflow and expectations of both user groups,

ensuring that the platform's design remains cantered around real user needs rather than purely technical considerations.

2.4.1 Employer

Employers using the platform primarily seek to identify, hire, and manage skilled freelancers for their specific project needs in a cost-effective and efficient manner. To support this, the platform must fulfil the following user requirements for employers:

- Ability to register and create a detailed employer profile, including company or individual information, to establish credibility with prospective freelancers.
- Ability to post projects with clearly defined requirements, including a description of the work, required skills, budget range, and expected timeline.
- Ability to search, filter, and shortlist freelancers based on skills, ratings, past work experience, and portfolio quality.
- Ability to review proposals submitted by freelancers, communicate with candidates, and select the most suitable one for a given project.
- Ability to monitor project progress through milestone tracking and communicate directly with the hired freelancer throughout the engagement.
- Ability to release payments securely through the escrow system upon satisfactory verification of completed milestones or final deliverables.
- Ability to rate and review freelancers upon project completion, contributing to the platform's overall trust and reputation system.

2.4.2 Freelancer

Freelancers using the platform are primarily focused on discovering relevant work opportunities, showcasing their skills effectively, and receiving fair and timely compensation for completed work. To support this, the platform must fulfil the following user requirements for freelancers:

- Ability to create and continuously update a detailed professional profile, including skills, work experience, educational background, certifications, and a portfolio of past projects.

- Ability to search and filter available projects based on category, budget, required skills, and timeline, with AI-assisted recommendations surfacing the most relevant opportunities.
- Ability to submit proposals or bids for projects, including a personalized cover message, proposed pricing, and estimated delivery timeline.
- Ability to communicate with employers regarding project requirements, clarifications, and progress updates through an in-platform messaging system.
- Ability to submit completed work against defined milestones and track the status of payment release through the escrow system.
- Ability to receive payments securely and promptly upon employer approval of delivered work, without unnecessary delays or excessive service fees.
- Ability to build a verifiable reputation over time through ratings and reviews received from employers after successful project completions.

2.5 Feasibility Analysis

Before proceeding with full-scale development, a feasibility analysis was conducted to assess whether the proposed freelancing platform could be realistically built, deployed, and adopted within the available resources, timeline, and technical constraints. This analysis was carried out across three key dimensions: economic, technical, and operational feasibility.

2.5.1 Economic Feasibility

The project is economically feasible primarily due to its reliance on open-source technologies and cost-effective cloud infrastructure, which significantly reduces both development and operational expenses. The entire technology stack, including React.js, Node.js, Express.js, and MongoDB, is freely available under open-source licenses, eliminating the need for expensive proprietary software or licensing fees. Additionally, cloud hosting providers such as AWS, Azure, or Render offer flexible, pay-as-you-go pricing models and free-tier options suitable for initial deployment and testing, allowing the project to scale its infrastructure costs in line with actual user growth rather than requiring substantial upfront investment. Third-party services required for payment processing, such as Razorpay or Stripe, similarly operate on a transaction-based fee model rather than fixed licensing costs, further keeping initial expenditure low. Given these factors, the project can be developed and initially deployed with minimal financial investment, making it well-suited for an academic or early-stage startup context.

2.5.2 Technical Feasibility

From a technical standpoint, the project is highly feasible, as it relies entirely on modern, well-documented, and widely-adopted web development frameworks and tools. React.js and Tailwind CSS provide a mature ecosystem for building responsive, component-based user interfaces, while Node.js and Express.js offer a proven, scalable foundation for backend API development. MongoDB's flexible, document-based data model is well-suited to handling the varied data structures involved in freelancer profiles, project listings, and transaction records. Furthermore, the AI-based recommendation functionality can be implemented using established machine learning libraries and pre-trained models, avoiding the need to develop complex algorithms from scratch. The extensive community support, comprehensive documentation, and abundance of learning resources available for each of these technologies significantly reduce technical risk and development time, confirming that the project is technically achievable within the given timeframe and skill set of the development team.

2.5.3 Operational Feasibility

Operationally, the platform has been designed with simplicity and ease of use as a core priority, ensuring that both freelancers and employers, regardless of their technical background, can navigate and utilize its features with minimal guidance or training. The intuitive dashboard layouts, straightforward project posting and bidding workflows, and clearly guided payment processes are designed to closely mirror user expectations shaped by familiarity with existing digital platforms, reducing the learning curve for new users. Additionally, since the platform is entirely web-based and does not require any specialized hardware or software installation beyond a standard browser and internet connection, it can be readily adopted by users across varying levels of digital literacy and technical infrastructure, further reinforcing its operational feasibility for widespread real-world use.

CHAPTER 3: SYSTEM DESIGN AND USE CASE DIAGRAMS

3.1 System Architecture

The MasterHire platform follows a three-tier client-server architecture, a widely adopted design pattern that separates the system into distinct layers: presentation, application logic, and data storage. This separation ensures that each layer can be developed, maintained, and scaled independently, while communicating with adjacent layers through well-defined interfaces.

At the presentation layer, the **frontend**, built using React.js and Tailwind CSS, is responsible for rendering the user interface and handling all client-side interactions, such as browsing jobs, submitting proposals, and viewing dashboards. When a user performs an action, the frontend sends requests to the **backend**, built using Node.js and Express.js, over a RESTful API using HTTPS. The backend handles core business logic, including authentication, job matching, negotiation, and milestone management, and communicates with the **database**, MongoDB, to store and retrieve data such as user profiles, job listings, proposals, and transaction records.

For payment-related operations, the backend additionally integrates with an external **payment gateway** (such as Razorpay or Stripe) to process wallet top-ups, escrow deposits, and withdrawal requests securely, without the platform directly handling sensitive card or banking information. This layered architecture ensures that the system remains modular, secure, and capable of scaling each component independently as the platform's user base grows.

MASTERHIRE PLATFORM - SERVICE-ORIENTED ARCHITECTURE

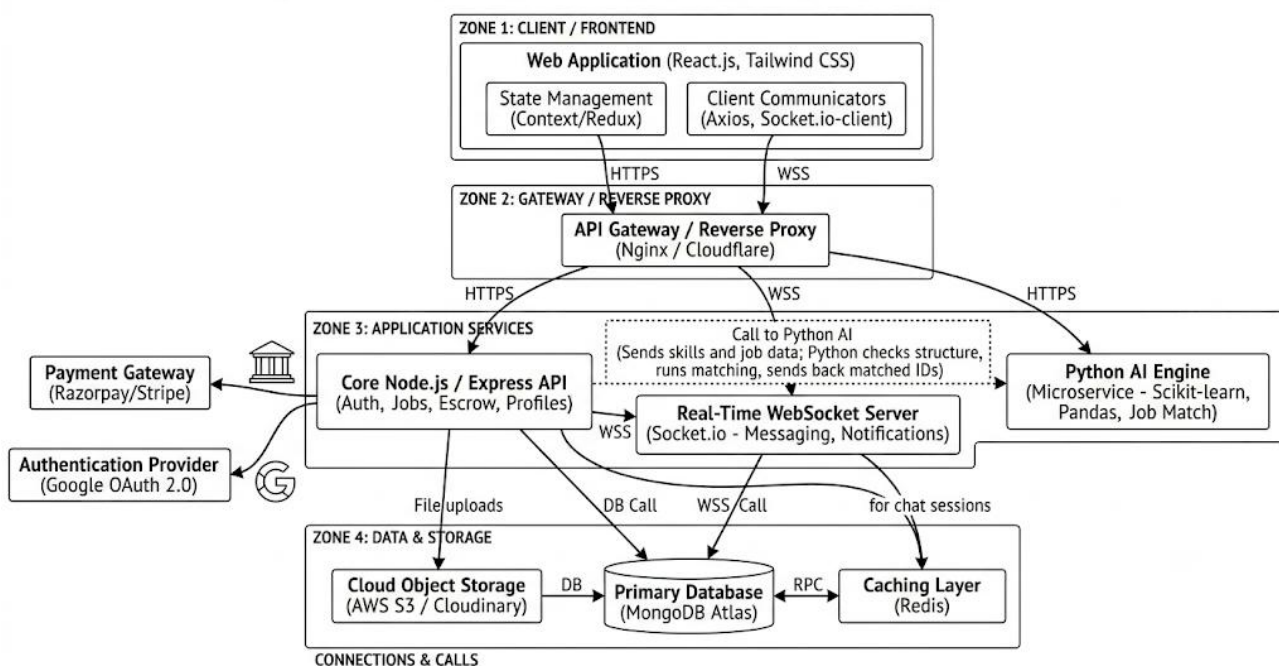


Fig. 3.1.A

3.2 Use Case Diagram

A use case diagram provides a high-level visual representation of the functional scope of a system, showing the different actors who interact with it and the specific actions, or use cases, each actor is able to perform. Unlike a flowchart, which captures the step-by-step logic of a single process, a use case diagram captures the overall capabilities of the system from the perspective of its users, making it a useful tool for understanding the full functional scope of MasterHire at a glance.

The MasterHire platform involves three primary actors: the **Client**, who posts projects and hires freelancers; the **Freelancer**, who searches for and completes work; and the **Admin**, who oversees platform-wide operations. The diagram below illustrates the key use cases associated with each actor, along with shared functionalities such as real-time chat and support access that are common to both Client and Freelancer roles.

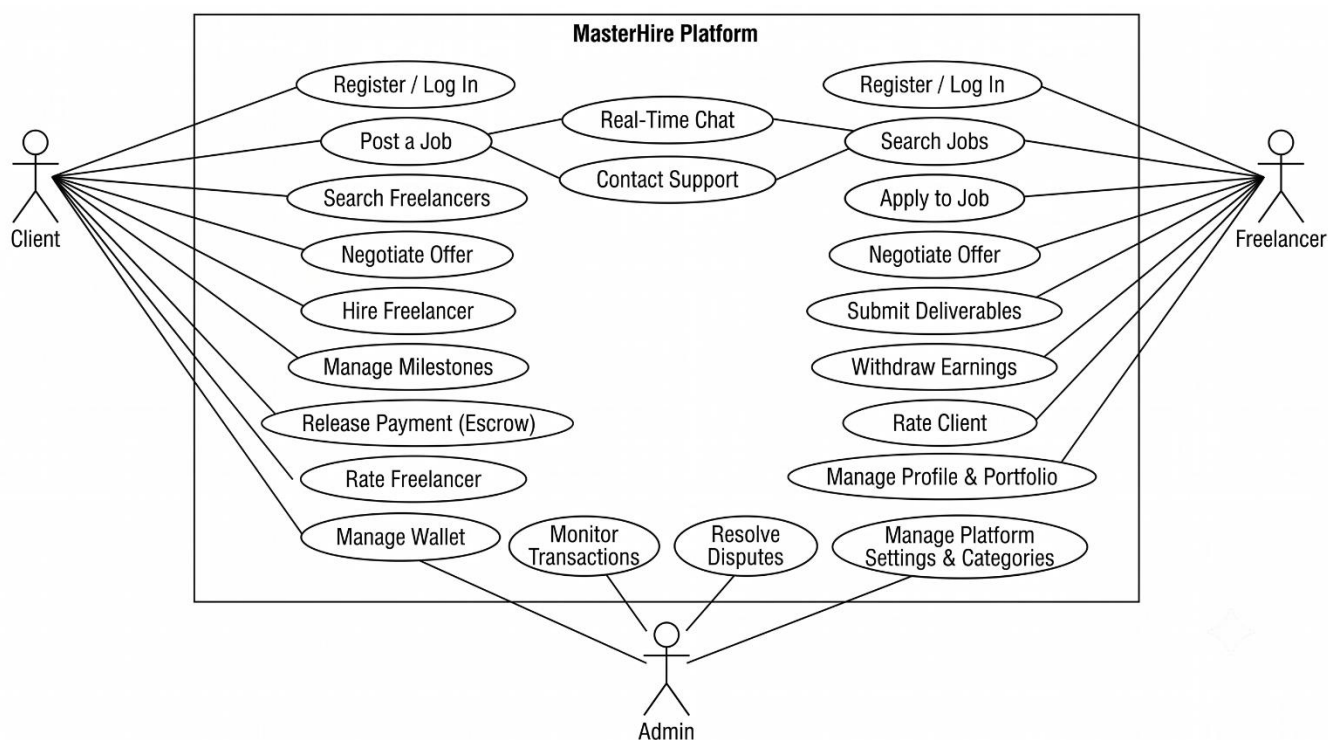


Fig. 3.2.A

As shown in the diagram, the Client's responsibilities span the full hiring lifecycle, from registering and posting a job, through negotiating and hiring a freelancer, to managing milestones and releasing escrow payments. The Freelancer's use cases mirror this from the opposite side, covering job discovery, application, deliverable submission, and withdrawal of earnings. The Admin actor, while not yet implemented with a dedicated interface in the current version of the platform, represents the intended scope for future administrative control, including user management, transaction monitoring, dispute resolution, and platform configuration.

3.3 Workflow Diagrams

3.3.1 User Registration and Role Selection Flow

The registration flow represents the entry point for every new user on the MasterHire platform, determining which dashboard and feature set they are directed to for the remainder of their experience. The process begins the moment a visitor chooses to sign up, and ends with the user landing on a role-specific dashboard tailored to either hiring or freelancing activities.

As shown in the flowchart, the user first selects their role, either Client or Freelancer, before proceeding to fill in their registration details or signing up instantly through Google authentication. Once the account is created, the system automatically redirects the user to the appropriate dashboard based on the role selected at the start of the process, ensuring that Clients and Freelancers immediately see the tools relevant to their respective needs.

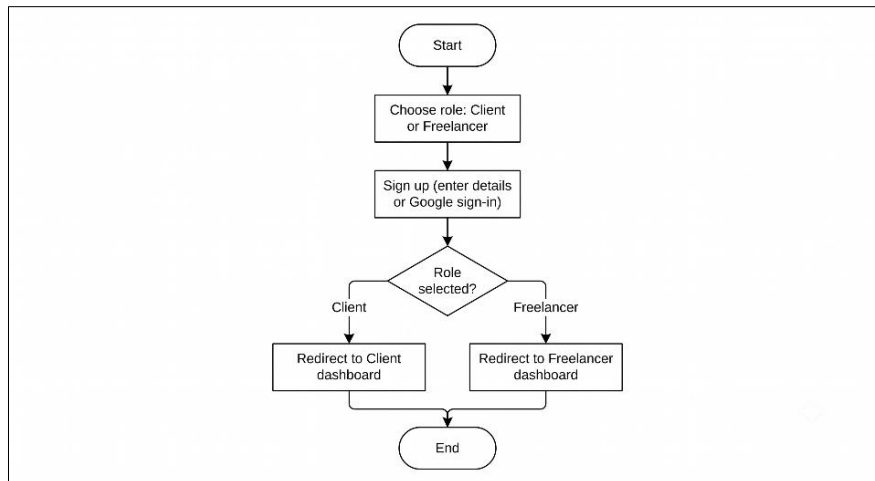


Fig. 3.3.1.A

3.3.2 Job Posting Flow

The job posting flow captures the sequence of steps a Client follows to publish a new project on the platform, from entering the initial project details through to validation and successful publication. This process is central to the platform's core functionality, as every subsequent interaction, from freelancer applications to negotiations and hiring, begins with a job being posted.

As illustrated in the flowchart, the Client begins by filling in the essential job details, including the job title, required skills, description, experience level, budget, deadline, etc. Upon submission, the system validates the entered information; if any required fields are missing or incorrectly filled, an error is shown and the Client is returned to the form to make corrections. Once the form passes validation, the job is published and becomes visible to freelancers browsing or searching for relevant opportunities.

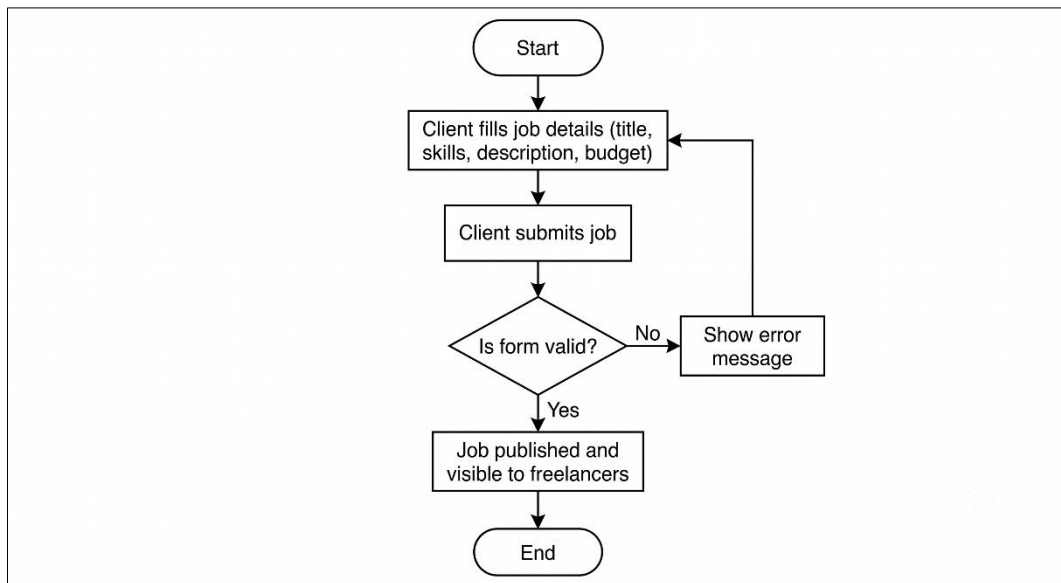


Fig. 3.3.2.A

3.3.3 Job Application and Negotiation Flow

The job application and negotiation flow describes how a Freelancer discovers and applies for a project, and how the Client and Freelancer arrive at a mutually agreed engagement through negotiation before work begins. This process is a critical trust-building step in the platform, ensuring that both parties explicitly agree on scope and pricing before any commitment is made.

As shown in the flowchart, the process begins with the Freelancer searching for and finding a suitable job listing, followed by submitting a proposal. The Client then reviews the received proposal, after which either party may initiate negotiation through offers and counter-offers exchanged via the platform's negotiation interface. If the offer is not accepted, negotiation continues or is ended by either party; if accepted, the Client proceeds to hire the Freelancer, formally starting the project engagement.

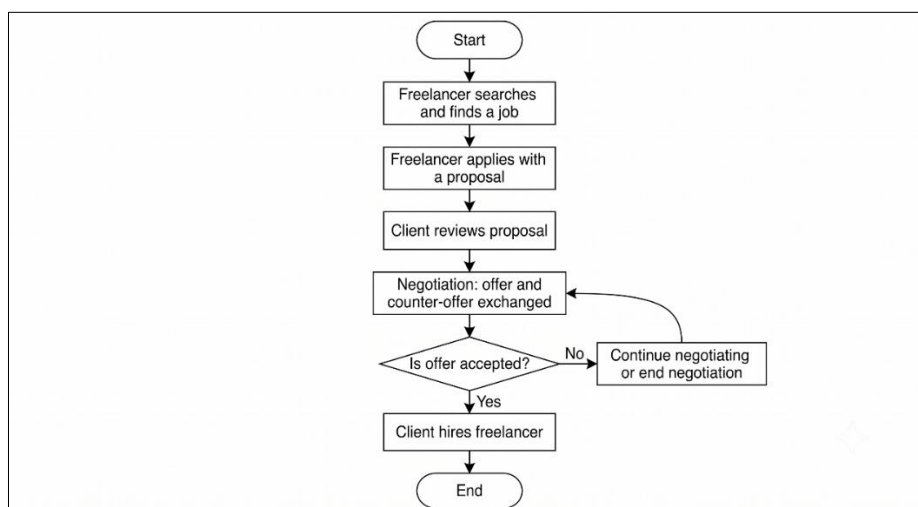


Fig. 3.3.3.A

3.3.4 Milestone and Payment Flow

The milestone and payment flow captures how project milestones are proposed, reviewed, and ultimately compensated once a Client has hired a Freelancer. This process is central to the platform's escrow-based trust model, ensuring that both parties agree on the distribution of work and payment before any deliverable submission begins, and that Freelancers are paid fairly once work is reviewed and approved.

As shown in the flowchart, the process begins with the Freelancer proposing a milestone breakdown, which is sent to the Client for review. The Client verifies the proposed milestones and either approves them or requests modifications to the milestone distribution; if modifications are requested, the Freelancer revises and resubmits the milestones until an agreement is reached. Once the milestones are approved, the Freelancer submits the corresponding deliverable, which the Client then reviews. Upon approval, the Client makes payment either from their existing wallet balance or directly through a payment method, after which the payment is transferred to the Freelancer following the deduction of applicable platform fees and taxes. The Freelancer may then withdraw their earnings from their wallet to their linked bank account or UPI ID, completing the milestone cycle.

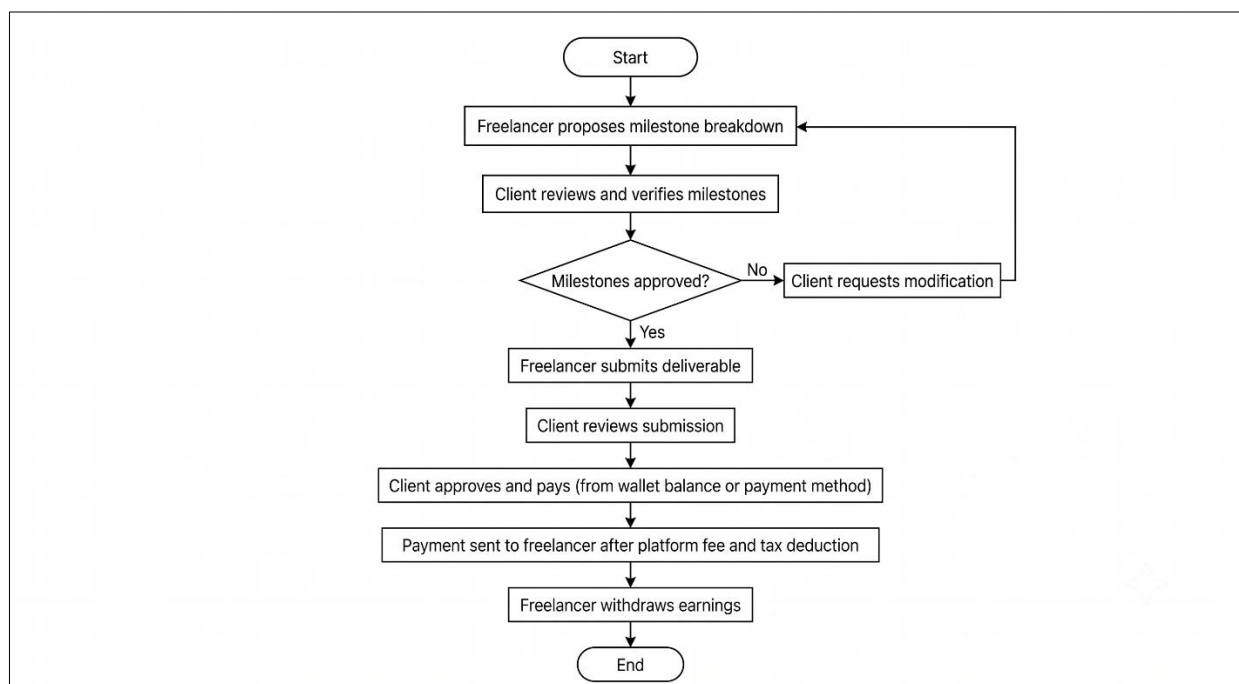


Fig. 3.3.4.A

CHAPTER 4: TOOLS AND TECHNOLOGIES USED

This chapter provides a detailed overview of the tools, frameworks, and technologies employed in the design, development, and deployment of the freelancing platform. The selection of each technology was guided by factors such as scalability, community support, development speed, and suitability for building a real-time, data-intensive web application.

4.1 Frontend Technologies

The frontend of the platform, responsible for all user-facing interactions, was built using **React.js**, a widely-adopted JavaScript library for building dynamic, component-based user interfaces. React's virtual DOM and component reusability make it particularly well-suited for a platform like this, where multiple user roles, namely freelancers, employers, and administrators, require distinct yet interconnected dashboards and views that update in real time as users interact with job postings, proposals, and messages. Its component-based structure also allows individual UI elements, such as project cards, profile forms, and rating widgets, to be developed, tested, and reused independently across different parts of the application.

Alongside React, **Tailwind CSS** was used as the primary styling framework. Tailwind's utility-first approach allows for rapid development of clean, modern, and fully responsive interfaces without the overhead of writing extensive custom CSS, ensuring that the platform renders consistently across desktops, tablets, and mobile devices. Together, React and Tailwind CSS enable the creation of an interface that is both visually polished and highly maintainable, allowing new features and pages to be added with minimal disruption to existing components. This combination also supports faster iteration during development, which is particularly valuable given the project's timeline constraints.

4.2 Backend Technologies

The backend of the platform is built using **Node.js** in conjunction with the **Express.js** framework, forming the core server-side logic that handles user authentication, job posting and bidding workflows, payment processing, and communication between the frontend and the database. Node.js was selected for its non-blocking, event-driven architecture, which is particularly well-suited to handling multiple concurrent user requests, such as simultaneous job applications or real-time notifications, without significant performance degradation.

Express.js, a minimal and flexible Node.js framework, is used to structure the backend using a **RESTful API architecture**, allowing the frontend to communicate with the server through clearly defined, stateless HTTP endpoints for operations such as user registration, project creation, proposal submission, and

payment initiation. This architecture ensures a clean separation of concerns between the frontend and backend, simplifying both development and future maintenance or scaling of the system. Middleware components within Express also handle cross-cutting concerns such as request validation, authentication token verification, and error handling in a centralized and consistent manner.

4.3 Database Technologies

MongoDB, a NoSQL, document-oriented database, serves as the primary data store for the platform. MongoDB's flexible, schema-less data model is particularly advantageous for a freelancing platform, where different types of records, such as freelancer profiles, project listings, proposals, and chat histories, often vary significantly in structure and may evolve over time as new features are introduced. This flexibility allows the development team to iterate on the data model without requiring complex database migrations, while MongoDB's native support for JSON-like documents (BSON) integrates naturally with the JavaScript-based Node.js backend, simplifying data handling across the stack.

Additionally, MongoDB's support for indexing and aggregation pipelines enables efficient querying and filtering, which is essential for features such as job search, freelancer discovery, and analytics dashboards. For deployment, MongoDB Atlas, a fully managed cloud database service, can be used to provide automated backups, scaling, and monitoring, reducing the operational overhead of managing database infrastructure manually.

4.4 AI and Analytics

To support intelligent job matching and data-driven decision-making, the platform incorporates AI and analytics components built primarily using **Python-based machine learning libraries**, which are integrated with the core Node.js backend through dedicated API services. The recommendation engine analyzes freelancer skills, past project history, and employer requirements to suggest the most relevant matches, using skill-similarity scoring and collaborative filtering techniques commonly applied in recommendation systems.

In addition to job matching, the analytics component tracks platform-wide metrics such as user engagement, project completion rates, and search trends, presenting these insights through visual dashboards for both administrators and, where relevant, employers seeking to understand candidate pools. This combination of AI-driven recommendations and analytical reporting allows the platform to continuously improve its matching accuracy over time, while also providing valuable business insights to guide future feature development and platform growth.

CHAPTER 5: RESULTS AND SCREENSHOTS

This chapter presents the practical outcome of the design and development phases discussed in the earlier chapters, showcasing the actual implemented interfaces of the MasterHire platform. The screenshots included in this chapter demonstrate how the core features, ranging from user authentication and job discovery to negotiation, communication, and payment management, have been translated into a functional, real-world application. Each screenshot is accompanied by a brief description explaining its purpose within the overall user journey.

5.1 Landing Page and Authentication

Landing Page: Highlights MasterHire's zero-fee value proposition, a dual-mode search for talent or jobs, 1,000+ browsable skill categories, client testimonials, and live platform statistics.

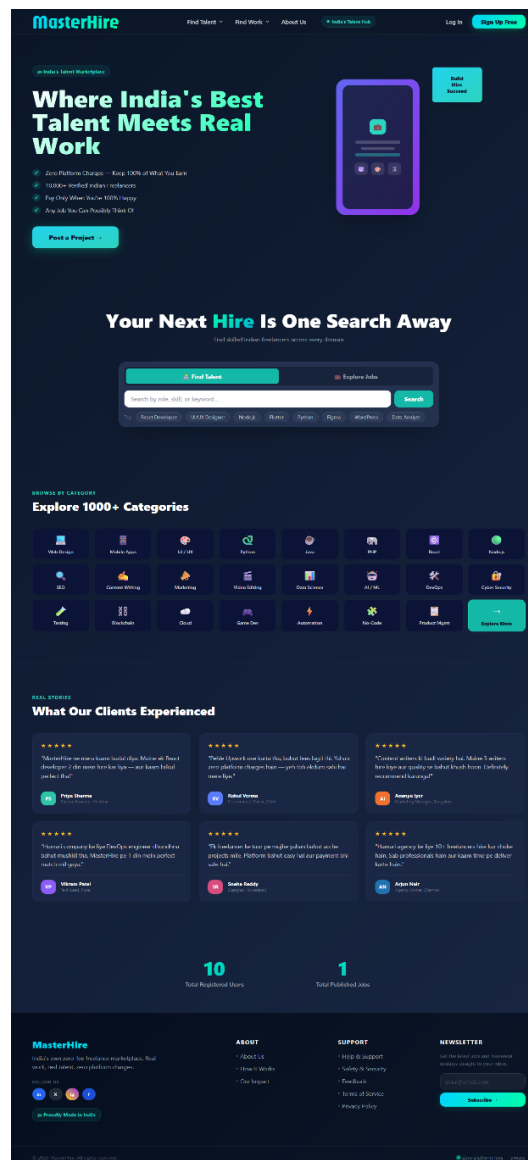
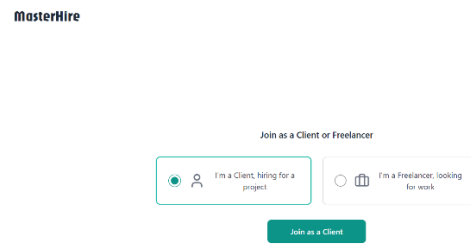


Fig. 5.1.A

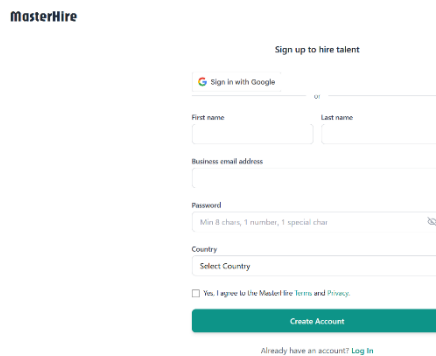
Choose Role Page: Lets a new user select their role, Client or Freelancer, which determines the dashboard and features they're directed to.



The image shows the 'Choose Role Page' for MasterHire. At the top left is the 'MasterHire' logo. Below it, the text 'Join as a Client or Freelancer' is centered. There are two main options: 'I'm a Client, hiring for a project' (selected with a green dot) and 'I'm a Freelancer, looking for work' (unselected with a grey dot). Below these options is a green button labeled 'Join as a Client'.

Fig. 5.1.B

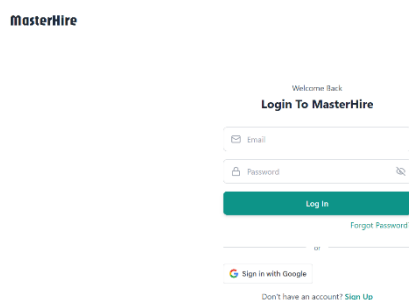
Sign-Up Page: The sign-up page allows a new user to create a MasterHire account either instantly via Google sign-in or by manually entering their basic details such as name, email address, password.



The image shows the 'Sign-Up Page' for MasterHire. At the top left is the 'MasterHire' logo. Below it, the text 'Sign up to hire talent' is centered. There are two main options: 'Sign in with Google' (selected with a green dot) and 'or' (unselected with a grey dot). Below these options are input fields for 'First name', 'Last name', 'Business email address', 'Password' (with a note 'Min 8 chars, 1 number, 1 special char'), and 'Country' (with a dropdown menu). Below these fields is a green button labeled 'Create Account'. At the bottom, there is a link 'Already have an account? Log In'.

Fig. 5.1.C

Login Page: Returning users authenticate through the login page by entering their registered email and password, or alternatively through one-click Google sign-in.



The image shows the 'Login Page' for MasterHire. At the top left is the 'MasterHire' logo. Below it, the text 'Welcome Back' is centered, followed by 'Login To MasterHire'. There are two main options: 'Email' (selected with a green dot) and 'or' (unselected with a grey dot). Below these options are input fields for 'Email' and 'Password' (with a note 'Min 8 chars, 1 number, 1 special char'). Below these fields is a green button labeled 'Log In'. At the bottom, there is a link 'Forgot Password?'. Below the 'Log In' button, there is a link 'Sign Up'.

Fig. 5.1.D

5.2 Employer (Client) Interface

Client Dashboard: Greet the client by name and displays a job overview across Assigned, Published, and Draft tabs, each linking to "Manage Work." It also surfaces recommended freelancer categories, top consultation providers, recently viewed profiles, and quick-access help resources.

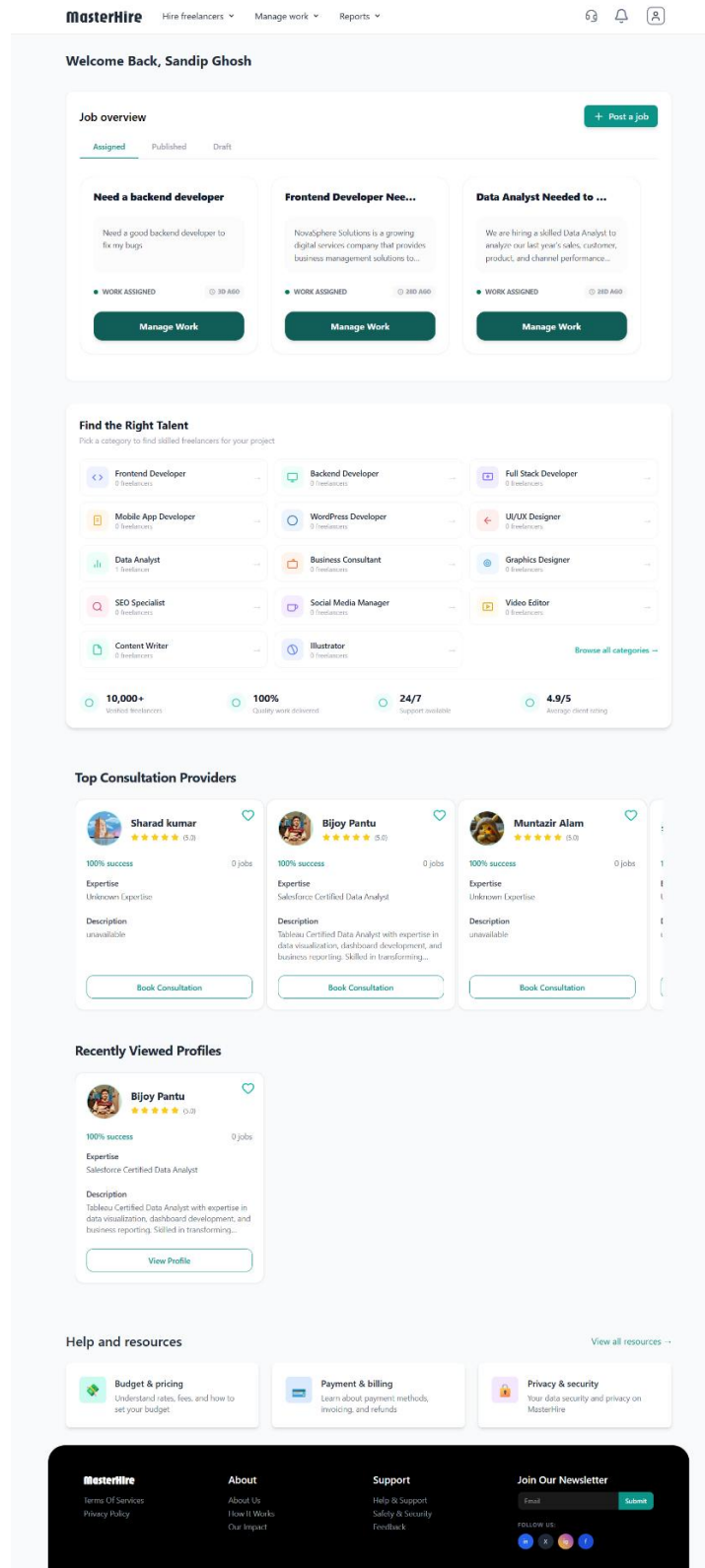


Fig. 5.2.A

Job Posting Page: A guided, section-by-section form where the client enters the job title, required skills, and a detailed project description, and other details in next steps with example prompts to help clients frame clear, well-scoped requirements before proceeding to the next step.

MasterHire Hire freelancers Manage work Reports

Basic Details

Job title

Write a clear, specific job title that describes what you want to get done.
A good title helps the right freelancers understand your project quickly.

Example:

- Build a Professional WordPress Website for My Business
- Develop a Custom Web Application Using Laravel
- Design and Build a Responsive Landing Page

Skills

List the specific skills your project requires.
Adding the right skills helps freelancers understand your expectations.

Job description

Clearly describe what you need done, including the project scope and key requirements.

Talents are looking for:

- Clear expectations about final deliverables
- Specific features you want included
- How you prefer to work (milestones, communication)

Attach files (pdf, doc, zip (max 10 MB))

Next

Fig. 5.2.B

Client Profile Page: Displays the client's company details, verification status, and key metrics (projects posted, freelancers hired, total spent, projects completed), alongside recent activity, freelancer reviews, and payment method management.

MasterHire Hire freelancers Manage work Reports

Sandip Ghosh (Email Verified) Edit Profile

Key Metrics:

- 4 Projects Posted
- 0 Freelancers Hired
- ₹0 Total Spent
- 3 Projects Completed
- 0 Repeat Hires

About Company Edit

Industry: [Blank]
Company Size: 1 - 10 Employees
Website: [Blank]
Member Since: June 2026

Reviews from Freelancers View All

No reviews yet

Recent Activity

- Posted a new project "digi" (3 days ago)
- Hired Bipin Paria for "Need a backend developer" (3 days ago)
- Hired Bipin Paria for "Frontend Developer Needed to Fix Bugs and Improve Functionality of Existing Business Website" (3 months ago)
- Hired Bipin Paria for "Data Analyst Needed to Analyze 1-Year Sales & Performance Data for My Company's Web Scraping Company" (3 months ago)

Payment Method

No payment method added

Manage Payment Methods

Fig. 5.2.C

Freelancer Search Page: Lets clients search freelancers by name, skill, title, or email, and refine results using filters for category, skills, rating, English level, job success, and availability, displaying each freelancer's rating, total jobs, and earnings.

MasterHire Hire freelancers Manage work Reports

Find Talent
Discover skilled freelancers for your next project.

Search by name, skill, title, or email...

Filters

CATEGORY: All SKILLS: All RATING: All ENGLISH: Any level Available Now Offer Consultations

JOB SUCCESS: Any job success OTHER LANGUAGE: All

6 freelancers found Showing 1-6 of 6

Sharad kumar
Unknown Title
Location Private
0 TOTAL JOBS 100% JOB SUCCESS ₹0 TOTAL EARNED 5.0 ★ RATING

Bijoy Pantu
Salesforce Certified Data Analyst
Howrah, West Bengal
0 TOTAL JOBS 100% JOB SUCCESS ₹0 TOTAL EARNED 5.0 ★ RATING
Excel Tableau Databases Figma Python Airflow Spark ETL Postgres Docker

Muntazir Alam
Unknown Title
Location Private
0 TOTAL JOBS 100% JOB SUCCESS ₹0 TOTAL EARNED 5.0 ★ RATING

SH sharad singh
Unknown Title
Location Private
0 TOTAL JOBS 100% JOB SUCCESS ₹0 TOTAL EARNED 5.0 ★ RATING

SH sharad kumar
Unknown Title
Location Private
0 TOTAL JOBS 100% JOB SUCCESS ₹0 TOTAL EARNED 5.0 ★ RATING

Roronoa Zoro
Unknown Title
Location Private
0 TOTAL JOBS 100% JOB SUCCESS ₹0 TOTAL EARNED 5.0 ★ RATING

MasterHire
Terms Of Services
Privacy Policy

About
About Us
How It Works
Our Impact

Support
Help & Support
Safety & Security
Feedback

Join Our Newsletter
Email Submit

FOLLOW US:
Facebook Twitter Instagram LinkedIn

Fig. 5.2.D

5.3 Freelancer Interface

Freelancer Profile Page: Displays the freelancer's headline, bio, location, verification badge, and key stats (total jobs, earnings, job success rate), along with detailed sections for languages, skills, education, work experience, certifications, and work history.

The screenshot displays a freelancer's profile on the MasterHire platform. The header includes the MasterHire logo and navigation links for 'Find work', 'Deliver work', and 'Reports'. User avatars and a notification bell with a '2' badge are also present.

The profile section features a profile picture, a headline 'Salesforce Certified Data Analyst', and a bio: 'Tableau Certified Data Analyst with expertise in data visualization, dashboard development, and business reporting. Skilled in transforming complex datasets into clear, actionable insights using Tableau, SQL, Excel, and Power BI. Experienced in data cleaning, analysis, and performance tracking, with a strong focus on helping businesses make data-driven decisions through impactful and interactive...'. The location is 'Bijoy Pantu' with a verification badge and 'Howrah, West Bengal'. Key stats show '0 Total jobs', 'Private Total earnings', and '0% Job success'.

The profile is divided into several sections:

- Languages:** English (Fluent), Hindi (Conversational), and Bengali (Native/Bilingual).
- Skills:** Excel, Tableau, Databases, Figma, Python, Airflow, Spark, ETL, Postgres, and Docker.
- Education:** Budge Budge Institute of Technology, Bachelors - Computer Science, 2026.
- Work History:** Completed (0), In Progress (0). Nothing to show.
- Work Experience:** MIS Executive, Arkca Trading Pvt Ltd, 2026-05-07 - Present.
- Certifications:** Google Data Analytics (Coursera), Issued: July 2025 - No Expiry.
- Other Experiences:** Add volunteer work, hackathons, awards and more.

The footer contains the MasterHire logo, links to Terms Of Services, Privacy Policy, About Us, How It Works, Our Impact, Support, Help & Support, Safety & Security, and Feedback. It also includes a 'Join Our Newsletter' section with an email input field and a 'Submit' button, and a 'FOLLOW US:' section with social media icons for LinkedIn, X, Instagram, and Facebook.

Fig. 5.3.A

Find Jobs (Job Search) Page: Lets freelancers search postings by title, skill, or Job ID, and narrow results using filters for category, experience level, price range, project length, and client history, with each listing showing skills required, budget, and deadline.

MasterHire Find work ▾ Deliver work ▾ Reports ▾

Find Jobs
Browse projects that match your skills and expertise.

Q Search by title, skill, keyword, or Job ID (e.g. JOB-345)...

Filters

CATEGORY All EXPERIENCE LEVEL All PRICE RANGES All PROJECT LENGTH All ☐ Need consultations

CLIENT HISTORY All

6 jobs found Showing 1-6 of 6

Posted just now **Python Developer to Automate Excel Reporting Process** JOB ID : JOB-690270

We manually generate weekly Excel reports from multiple CSV files and want to automate the entire workflow. The freelancer should develop a Python script that cleans the data, merges files, performs calculations, generates formatted Excel reports, and logs errors. The solution should be well-documented and easy to maintain.

Skills: Python Pandas OpenPyXL Automation Microsoft Excel

Est. budget: ₹19000 Est. time: 2026-08-06 Proposals: ...

Posted 1min ago **Graphic Designer Needed for Monthly Social Media Content** JOB ID : JOB-175567

We are seeking a creative graphic designer to create social media posts, promotional banners, and marketing creatives for our brand. Approximately 25-30 designs will be required every month. The designer should maintain consistent branding and deliver high-quality editable source files.

Skills: Adobe Photoshop Adobe Illustrator Canva Graphic Design Branding

Est. budget: ₹10000 Est. time: 2026-08-12 Proposals: ...

Posted 2min ago **Redesign Existing WordPress Website for Better Performance** JOB ID : JOB-241737

Our company website requires a complete redesign with improved navigation, faster loading speed, and SEO optimization. The freelancer should migrate the existing content, improve the design, and ensure compatibility across all devices. Basic on-page SEO and security improvements are expected.

Skills: WordPress Elementor PHP CSS SEO Website Optimization

Est. budget: ₹20000 Est. time: 2026-08-06 Proposals: ...

Posted 2min ago **Full Stack Developer Required for Company Website Development** JOB ID : JOB-862758

We need a responsive business website with user authentication, an admin dashboard, and contact management features. The website should be optimized for desktop and mobile devices. The developer should follow modern UI/UX practices and deploy the application on a cloud platform. Previous portfolio work is preferred.

Skills: HTML CSS JavaScript React.js Node.js Express.js MongoDB REST API

Est. budget: ₹45000 Est. time: 2026-09-10 Proposals: ...

Posted 3min ago **Data Analyst Needed to Build Interactive Sales Dashboard** JOB ID : JOB-160634

We are looking for an experienced Data Analyst to analyze our e-commerce sales data from the last 12 months. The selected freelancer will clean the dataset, identify sales trends, calculate KPIs, and create an interactive Tableau dashboard. Experience with SQL and business reporting is required. The final deliverables should include the dashboard, cleaned dataset, and documentation...

Skills: SQL Microsoft Excel Tableau Data Analysis Data Visualization

Est. budget: ₹15000 Est. time: 2026-07-31 Proposals: ...

Posted 3d ago **dgds** JOB ID : JOB-327755

gggrldg

Skills: a b c

Est. budget: ₹ Est. time: 2026-07-31 Proposals: ...

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Fig. 5.3.B

Project History Page: Summarizes a freelancer's overall track record through total, completed, in-progress, and cancelled project counts alongside average rating, with individual project cards showing client, timeline, agreed price, completion percentage, and quick access to details or files.

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3
Total Projects
All time

0
Completed
0.0% success rate

3
In Progress
Currently active

0
Cancelled
Not completed

—
Avg. Rating
From 0 reviews

All Projects

Completed

In Progress

Cancelled

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Need a backend developer In Progress
Client: Client Beginner
Jun 12, 2026 – Ongoing | 3 days
next.js backend laravel oops +1
Need a good backend developer to fix my bugs
60% Completed
next.js backend + 2 more
₹9,500 Agreed Price
Awaiting review
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Frontend Developer Needed to Fix Bugs and Improve Functionality of Existing Business Website In Progress
Client: Client Intermediate
Jun 17, 2026 – Ongoing | 28 days
HTML5 CSS3 JavaScript React.js +8
NovaSphere Solutions is a growing digital services company that provides business...
60% Completed
HTML5 CSS3 + 2 more
₹25,000 Agreed Price
Awaiting review
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Data Analyst Needed to Analyze 1-Year Sales & Performance Data for My Company Velvet Scoop Creamery In Progress
Client: Client Beginner
Jun 17, 2026 – Ongoing | 28 days
Excel Google Sheets SQL Tableau +1
We are hiring a skilled Data Analyst to analyze our last year's sales, customer, product, and...
60% Completed
Excel Google Sheets + 2 more
₹16,000 Agreed Price
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Fig. 5.3.C

5.4 Project Application, Negotiation, and Communication

This section illustrates the complete lifecycle of a project engagement on MasterHire, from a freelancer applying to a listed job through to negotiation, active collaboration, file exchange, and milestone-based payment tracking.

Project Details and Apply Page: Displays the full job description, budget, experience level, required skills, and activity metrics such as proposals received and last client view, with clear "Apply to the job" and "Save job" actions for freelancers.

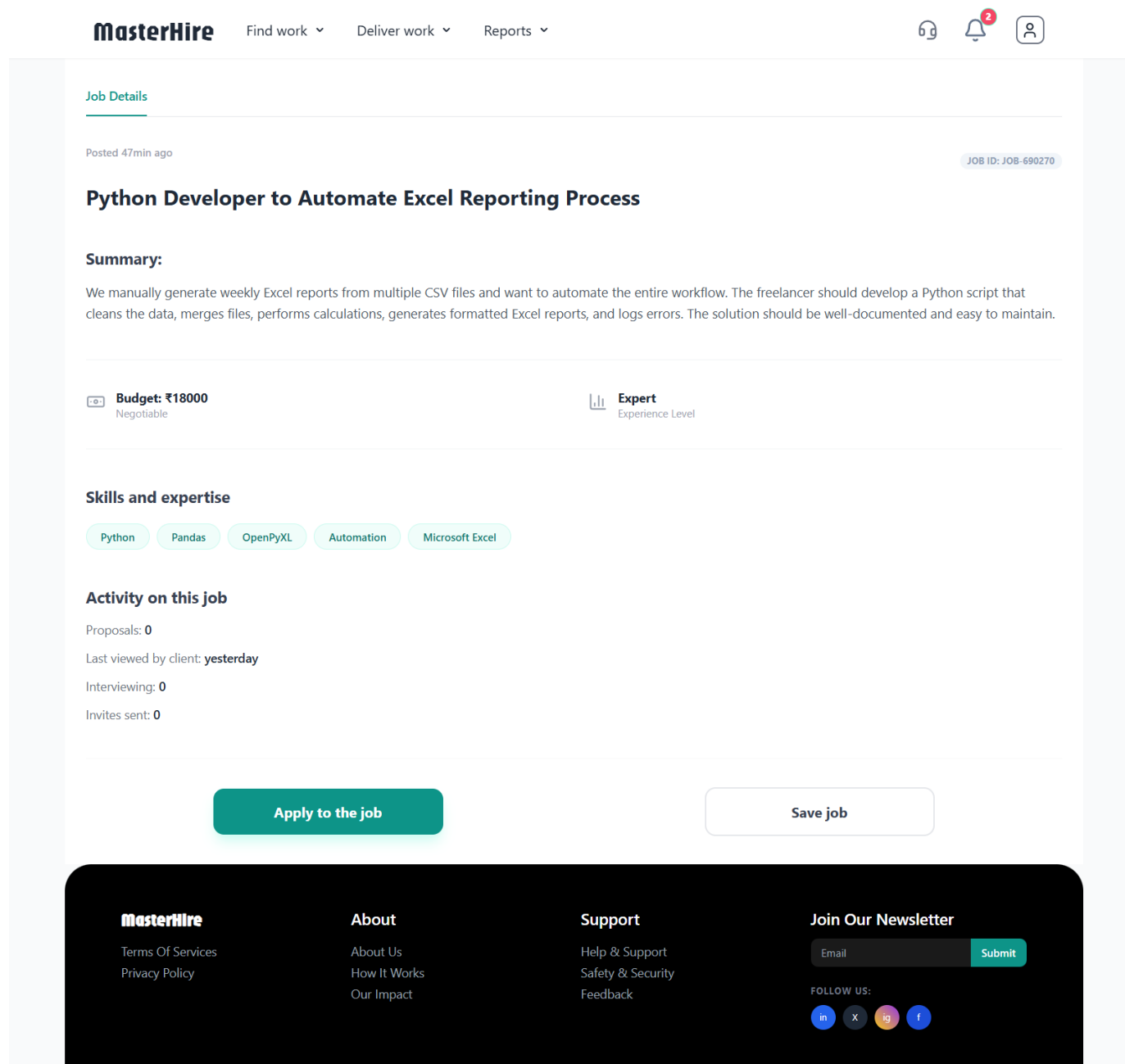


Fig. 5.4.A

Negotiation Page: Shows the back-and-forth bid history between client and freelancer, alongside a "Send Counter Offer" form and options to either "End Negotiation" or "Accept & Hire," ensuring the final agreed amount is explicitly recorded before work begins.

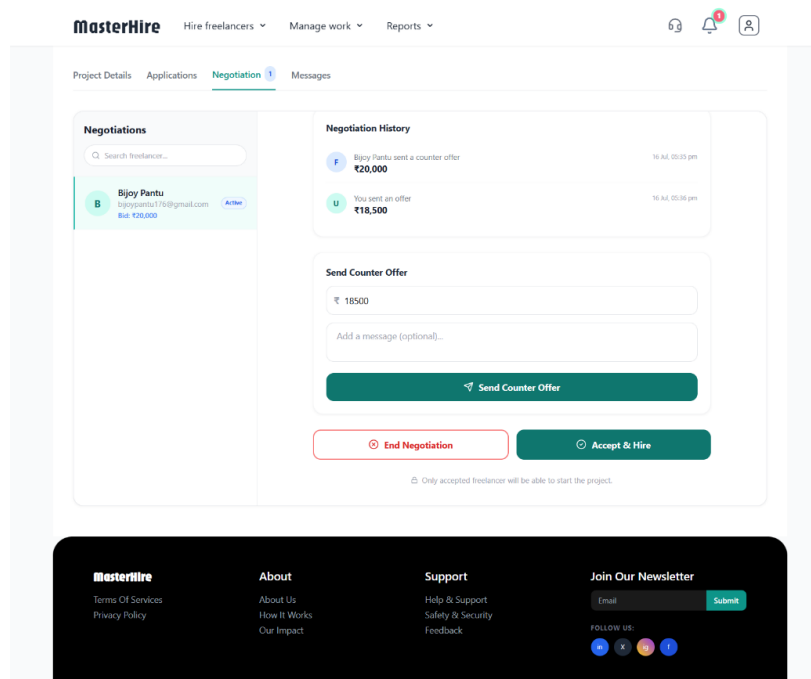


Fig. 5.4.B

Milestones Page: Summarizes the total contract value, amount earned, and pending payment, alongside individual milestones with due dates and amounts, an activity timeline of milestone proposals, and an earnings summary for tracking payment progress.

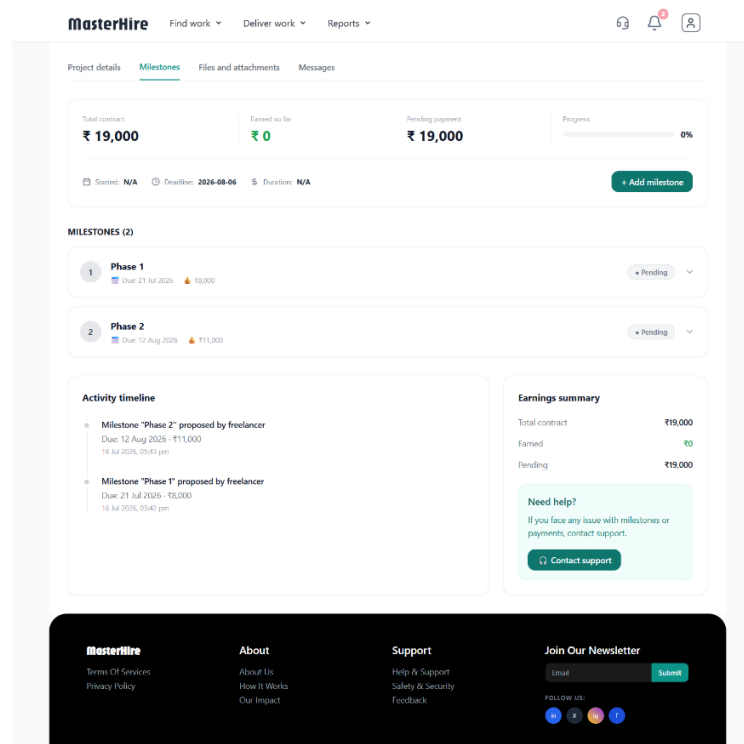


Fig. 5.4.C

Files and Attachments Page: Provides a drag-and-drop upload area for freelancers to submit deliverables for client review, alongside a separate section listing files shared by the client relevant to the project.

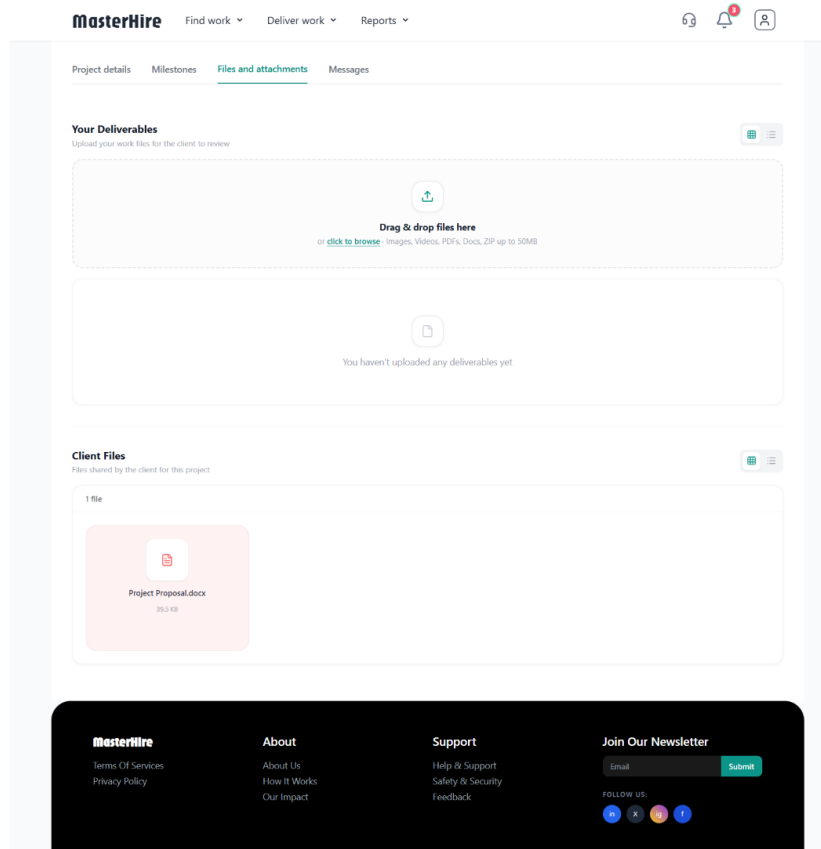


Fig. 5.4.D

Chat System: Enables real-time, project-specific messaging between client and freelancer, supporting text conversations, call and video-call options, and file/image sharing directly within the same workspace.

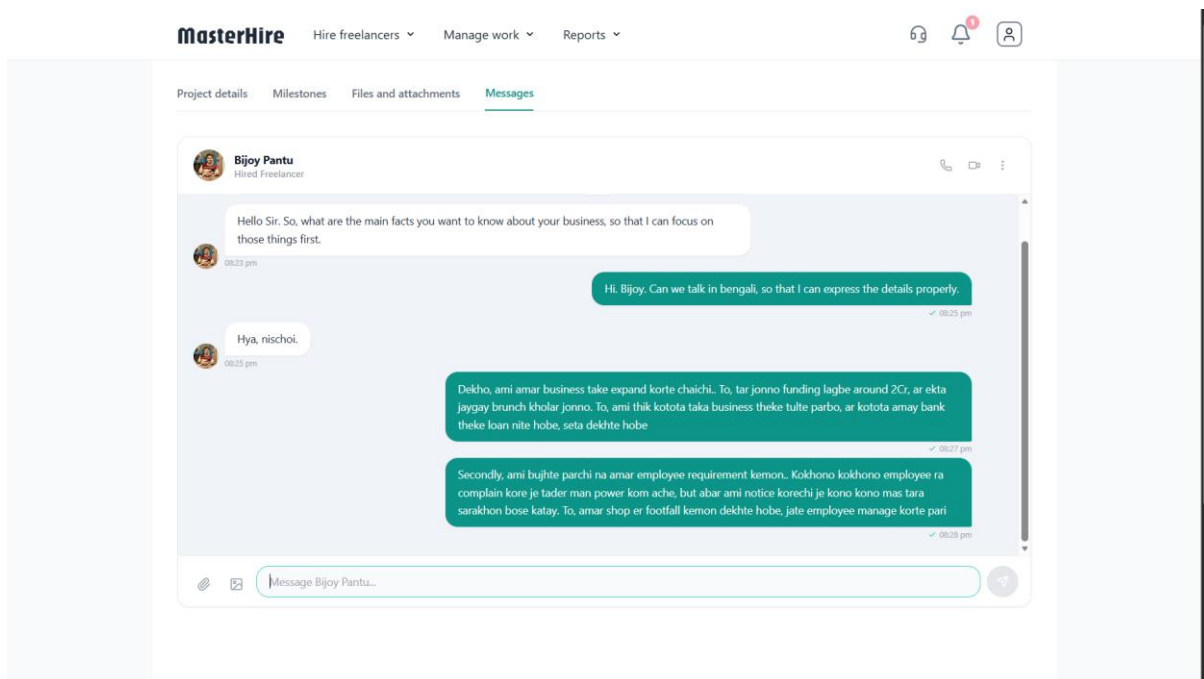


Fig. 5.4.E

5.5 Payments and Financial Dashboard

Client Wallet (Billing & Payments): Displays the client's current wallet balance with an "Add Funds" option, saved payment method and UPI details, billing address, and a payment history log recording each top-up transaction and its status.

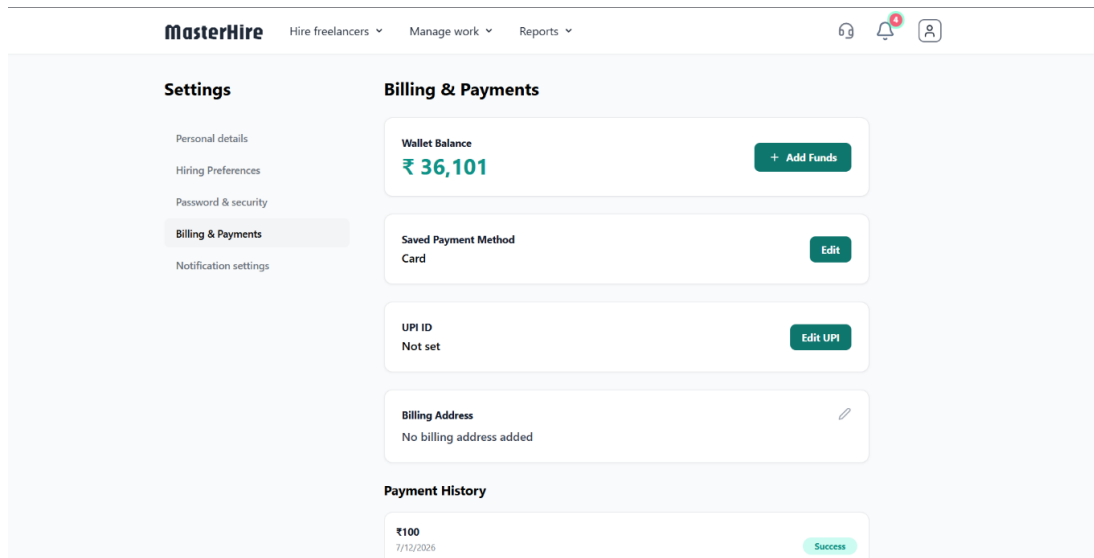


Fig. 5.5.A

Freelancer Wallet (Withdrawals): Shows the freelancer's withdrawal method (UPI ID), withdrawable balance, total earned versus total withdrawn, and wallet expiry notice, along with a "Withdraw Money" action and a log of past withdrawal requests.

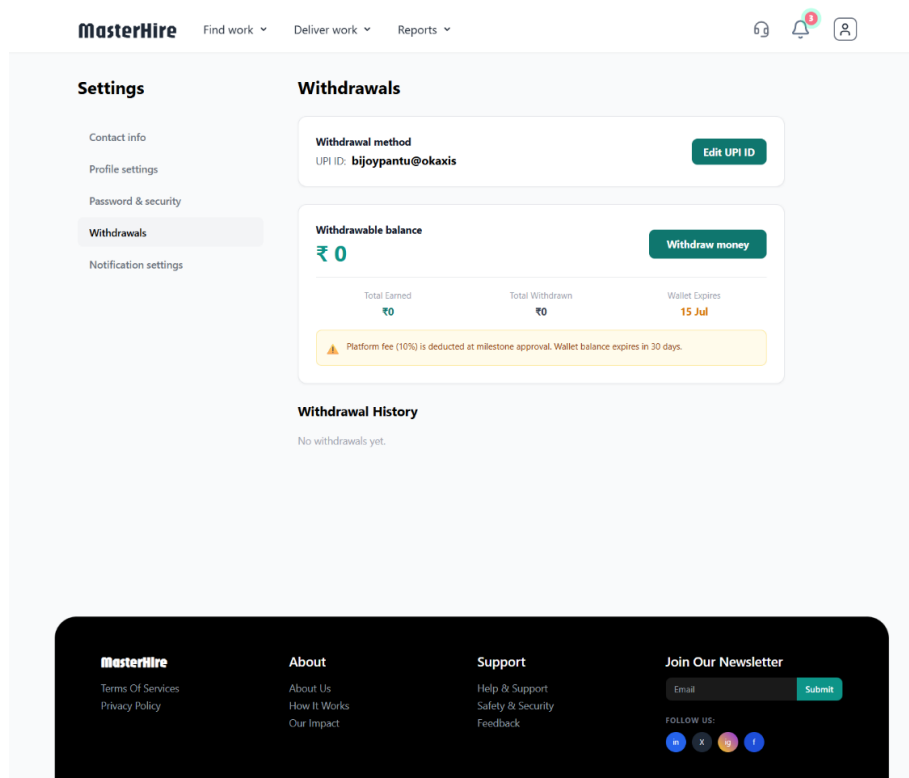
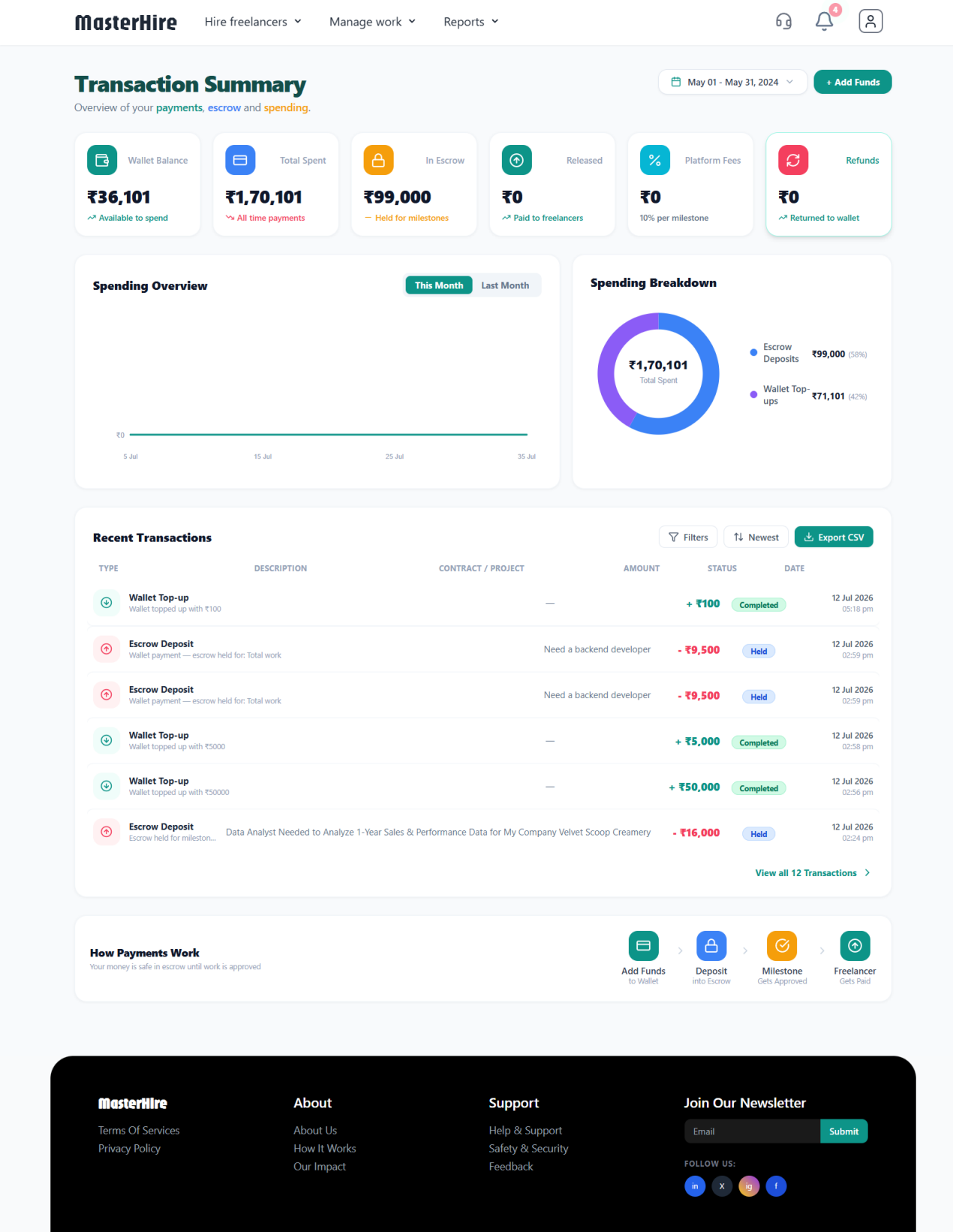


Fig. 5.5.B

Client Transaction Summary: Provides a consolidated view of wallet balance, total spent, funds held in escrow, amounts released to freelancers, platform fees, and refunds, supported by a spending overview graph, a spending breakdown chart, and a detailed, filterable list of recent transactions.



5.6 Settings and Support

Client Settings: The client-side settings panel includes Personal Details, Hiring Preferences, Password & Security, Billing & Payments, and Notification Settings. The Personal Details view shown here lets the client edit their name, email, company name, phone, location, and bio, with an account closure option at the bottom.

The screenshot shows the 'MasterHire' client settings interface. The top navigation bar includes 'MasterHire', 'Hire freelancers', 'Manage work', and 'Reports'. The left sidebar lists 'Settings' with sub-options: 'Personal details', 'Hiring Preferences', 'Password & security', 'Billing & Payments', and 'Notification settings'. The main content area is titled 'Personal Details' and features a profile card with the initials 'SG'. Below the card are input fields for 'Name' (Sandip Ghosh), 'Email' (bijoyarkca@gmail.com, marked as 'Primary'), 'Company Name', 'Phone', 'Location', and 'Bio'. At the bottom, there is a red 'Account closure' warning and a 'Close my account' button. The footer contains links for 'MasterHire', 'About', 'Support', and a 'Join Our Newsletter' form.

Fig. 5.6.A

Freelancer Settings: The freelancer-side settings panel includes Contact Info, Profile Settings, Password & Security, Withdrawals, and Notification Settings. The Profile Settings view shown here lets the freelancer manage their category and sub-specializations, experience level, profile visibility, and earnings privacy.

The screenshot shows the 'MasterHire' freelancer settings interface. The top navigation bar includes 'MasterHire', 'Find work', 'Deliver work', and 'Reports'. The left sidebar lists 'Settings' with sub-options: 'Contact info', 'Profile settings', 'Password & security', 'Withdrawals', and 'Notification settings'. The main content area is titled 'Profile settings' and includes a 'Category' section with 'Data Science & Analytics' and sub-specializations 'Data Analysis & Reporting', 'Data Extraction/ETL', and 'Data Mining & Management'. Below this is the 'Experience level' section with three options: 'Entry level' (selected), 'Intermediate', and 'Expert'. The 'Visibility' section has a dropdown menu set to 'Public'. The 'Earnings privacy' section has a toggle switch set to 'Private'. At the bottom, there are 'Save Changes' and 'Cancel' buttons. The footer contains links for 'MasterHire', 'About', 'Support', and a 'Join Our Newsletter' form.

Fig. 5.6.B

Notification Settings: A shared settings page allowing users to toggle notification preferences across Hiring & Proposals, Messages, Billing & Payments, and Marketing & Updates, giving fine-grained control over which platform alerts they receive.

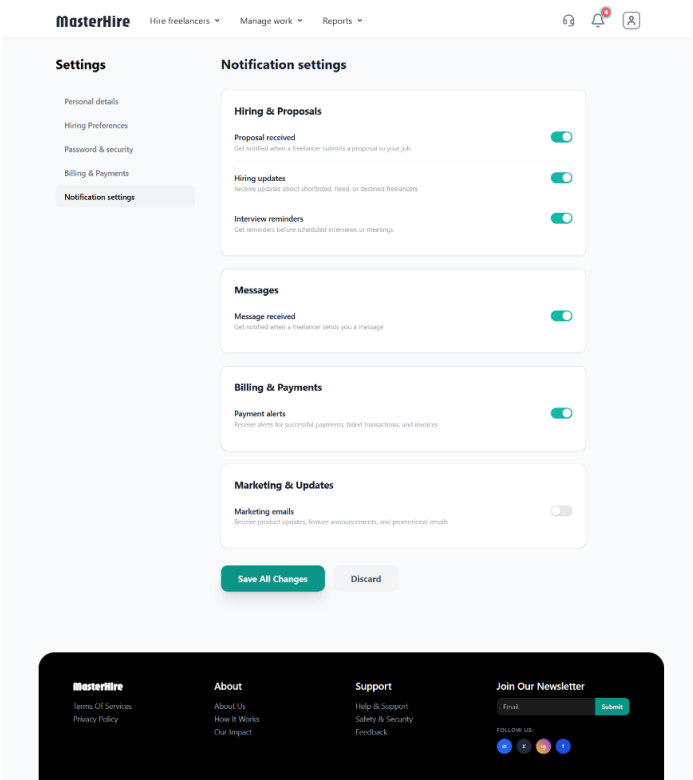


Fig. 5.6.C

Support System: A dedicated help center offering searchable articles across Getting Started, Payments, Safety & Trust, and Work Resources, along with direct "Start Live Chat" and "Submit a Ticket" options for unresolved issues.

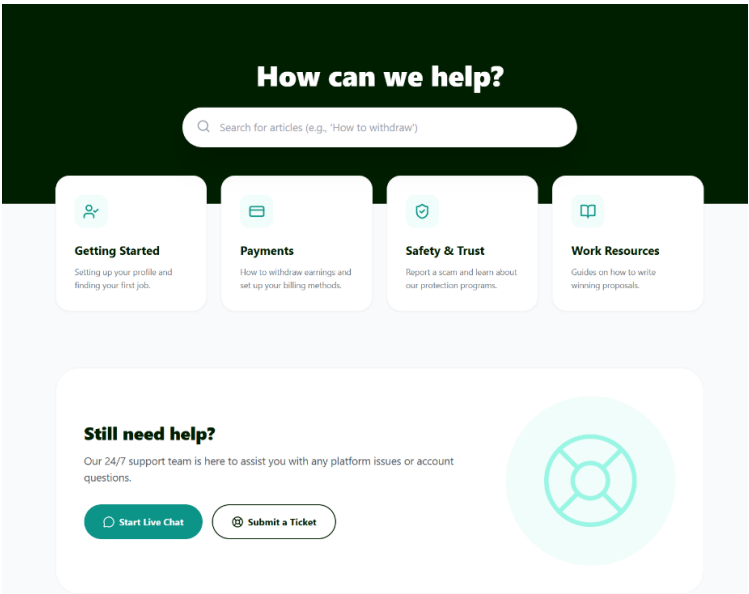


Fig. 5.6.D

CHAPTER 6: TESTING AND DEBUGGING

6.1 Purpose of Testing

The purpose of software testing is to evaluate the capabilities and attributes of a software program's ability to adequately meet applicable standards and application needs. Testing does not by itself ensure quality, nor is finding bugs its sole purpose; rather, testing serves as a means of verification, validation, and reliability estimation for the system as a whole. The primary objectives of testing in this project include identifying defects within the application before deployment, providing the development team with clear and actionable information about the system's current state, and verifying the correct functioning of core operations such as inserting, updating, and deleting records related to user profiles, projects, and transactions.

6.2 Types of Testing

Several standard types of testing were applied throughout the development of the platform to ensure comprehensive coverage of both individual components and the system as a whole:

- **Unit Testing** - Testing individual functions and components, such as form validation logic or API route handlers, in isolation to confirm they behave as expected.
- **Integration Testing** - Verifying that different modules, such as the frontend, backend API, and database, work correctly together, particularly around data flow between the project posting, bidding, and payment modules.
- **System Testing** - Testing the complete, integrated application as a whole to ensure it meets the specified functional and non-functional requirements.
- **User Acceptance Testing (UAT)** - Having sample users, acting as freelancers and employers, use the platform to confirm it meets real-world expectations before final submission.

6.3 Test Cases and Results

The following table presents a representative sample of test cases executed on the platform, covering core functionalities across user registration, project management, and payment workflows:

Test Case ID	Module	Test Description	Expected Result	Actual Result	Status
TC01	Authentication	Register with valid details	Account created, redirected to dashboard	Account created successfully	Pass
TC02	Authentication	Login with incorrect password	Error message shown, access denied	Error displayed correctly	Pass
TC03	Project Posting	Employer posts a new project with valid details	Project listed and visible to freelancers	Project posted successfully	Pass
TC04	Project Posting	Auto-redirect to dashboard after successful post	Redirects to dashboard automatically	No redirection occurred	Fail (Fixed)
TC05	Payment Settings	Client opens payment settings page	Page loads with payment options	Blank white page displayed	Fail (Fixed)
TC06	Project Workspace	Final negotiated amount reflects in workspace	Workspace shows finalized amount	Showed original proposed budget	Fail (Fixed)

TC07	Milestones	Sum of milestone amounts entered	Should equal total proposed amount	No validation enforced, mismatched totals allowed	Fail (Fixed)
TC08	Project Details Page	Display of amount, deadline, and details	Properly aligned within layout	Elements misaligned/shifted to one side	Fail (Fixed)
TC09	Escrow Payment	Release payment on milestone approval	Funds released to freelancer	Released correctly	Pass
TC10	Rating System	Submit rating after project completion	Rating saved and reflected on profile	Rating saved successfully	Pass

6.4 Tools Used for Testing

Testing was carried out using a combination of manual and automated approaches. **Postman** was used extensively for testing backend API endpoints independently of the frontend, allowing verification of request/response behavior for authentication, project, and payment routes. Browser developer tools (Chrome DevTools) were used for inspecting frontend rendering issues, network requests, and console errors during UI testing. Manual testing across multiple browsers and screen sizes was also carried out to catch layout and responsiveness issues, such as those identified in the project details page.

6.5 Bugs Identified and Resolved

During the testing phase, several bugs were identified through manual testing and user feedback, and subsequently resolved before final deployment. The key issues encountered are summarized below:

1. **Blank Payment Settings Page:** The payment settings page for clients was rendering as a completely blank white screen instead of displaying the payment configuration options. This was traced to an unhandled rendering error in the component and was resolved by fixing the underlying component logic and adding proper error handling.

2. **Incorrect Final Amount in Project Workspace:** After a client and freelancer negotiated and finalized a project amount different from the initial proposal, the project workspace continued to display the original proposed budget instead of the finalized negotiated amount. This was fixed by ensuring the workspace correctly fetched and displayed the updated, finalized amount rather than the initial one.
3. **Missing Milestone Amount Validation:** The milestone creation page did not enforce a check ensuring that the sum of all individual milestone amounts matched the total proposed project amount, allowing mismatched or inconsistent milestone totals to be submitted. A validation rule was added to ensure milestone amounts always sum up correctly to the total project amount.
4. **Misaligned Project Details Layout:** On the project details page, key information such as the amount, deadline, and other project details were not placed correctly within the layout and appeared shifted to one side instead of being properly aligned. This was corrected by fixing the layout structure and styling to ensure all details are displayed clearly and consistently.
5. **Failed Auto-Redirection After Project Posting:** After an employer successfully posted a new project, the platform was expected to automatically redirect them to their dashboard, but this redirection was not occurring. The issue was resolved by correctly implementing the post-submission navigation logic to ensure users are redirected as intended.

Addressing these issues during the testing phase significantly improved the reliability, usability, and overall polish of the platform ahead of final submission.

CHAPTER 7: CONCLUSION, LIMITATIONS, AND LESSONS

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7.1 Conclusion

This project set out to design and develop a centralized, secure, and intelligent freelancing platform tailored specifically to the needs of the Indian gig economy, addressing the notable gaps left by existing global platforms such as Upwork, Fiverr, and Freelancer.com. Through a systematic process of background research, requirement analysis, system design, and iterative development, the resulting platform successfully brings together freelancers and employers within a single, structured ecosystem that supports the entire engagement lifecycle, from project discovery and negotiation to final delivery and payment settlement.

The platform, built using a modern technology stack comprising React.js and Tailwind CSS on the frontend, Node.js and Express.js on the backend, and MongoDB as the underlying database, provides a responsive, scalable, and maintainable foundation capable of supporting a growing user base. Core functionalities such as detailed freelancer profile and portfolio management, structured project posting and bidding, milestone-based project tracking, and secure escrow-based payment processing have been implemented to directly address the trust, transparency, and payment-related concerns that were identified as major pain points during the initial literature review and background research phases. In addition, the integration of AI-based job recommendations and an analytics-driven search system helps reduce the friction typically associated with job discovery, allowing freelancers to more easily find relevant opportunities and employers to more efficiently identify suitable candidates.

The development process also placed significant emphasis on real-world reliability, as reflected in the dedicated testing and debugging phase carried out prior to final submission. Through a combination of manual testing, API-level testing using tools such as Postman, and user-acceptance style validation, several critical issues were identified and resolved, including a non-functional payment settings page, incorrect reflection of finalized project amounts within the project workspace, missing validation on milestone amount totals, layout inconsistencies on the project details page, and a broken auto-redirection flow after project posting. Addressing these issues before deployment significantly improved the platform's overall stability, usability, and readiness for real-world use, demonstrating that the system is not merely a theoretical design but a functioning application that has been rigorously tested against practical usage scenarios.

From a broader perspective, this project demonstrates the practical feasibility and value of building a localized alternative to global freelancing platforms, one that is specifically designed to address the financial, competitive, and trust-related challenges faced by Indian freelancers and employers. By offering features such as escrow-protected payments, transparent milestone tracking, and intelligent job matching, the platform aims to make freelancing a more accessible, reliable, and rewarding career path for a growing segment of India's digitally skilled workforce, while simultaneously providing employers with a dependable avenue to source project-based talent efficiently. In doing so, the project contributes, in a small but meaningful way, to the broader growth and maturation of India's gig economy.

While the current implementation successfully fulfils the core objectives outlined at the start of this project, it also represents a strong foundation upon which further enhancements, discussed in the following sections, can be built to expand the platform's capabilities and reach in future iterations.

7.2 Limitations

While the freelancing platform successfully implements its core objectives, certain limitations remain in the current version of the system, largely stemming from the scope and timeframe of an academic final-year project rather than fundamental flaws in the underlying approach.

Firstly, the platform is entirely web-based and currently lacks native mobile applications for iOS and Android, which limits accessibility for users who prefer conducting freelance work and communication primarily through mobile devices. Secondly, the payment infrastructure, while functional through escrow-based fund holding and milestone release, does not yet include direct UPI integration or automated TDS (Tax Deducted at Source) compliance and documentation, both of which are commonly expected in a payment system built specifically for the Indian market, and would require dedicated integration with banking or fintech APIs beyond the current implementation.

Thirdly, the AI-based job recommendation system, although functional, is still in an early stage in terms of matching accuracy, as it has not been trained or refined on a large volume of real-world usage data, meaning its recommendations may not yet be as precise as those seen in more mature, established platforms with years of accumulated user interaction data. Similarly, the platform currently lacks a dedicated dispute resolution mechanism for handling disagreements between freelancers and employers over deliverables, deadlines, or payment amounts, relying instead on direct communication through the in-platform chat system.

The platform is also currently limited to English as its primary interface language, without multi-language support that could improve accessibility for users in Tier 2 and Tier 3 cities who may be more comfortable

operating in regional languages. Additionally, since the platform is still in its testing and academic evaluation phase, it has not yet been deployed or stress-tested under real, large-scale user traffic, meaning its performance, scalability, and reliability under high concurrent load remain unverified in a production environment. Finally, features such as the wallet balance expiry period and platform fee structure, while functional, are currently based on fixed, hardcoded values rather than configurable business rules, which would need to be revisited before any real-world commercial deployment.

These limitations do not undermine the core value or functionality of the platform but instead highlight clear, well-defined areas for improvement, several of which are addressed in the Future Scope discussed in the following chapter.

7.3 Lessons Learnt

The development of this freelancing platform served as a valuable learning experience, extending well beyond the theoretical concepts studied in the classroom and into the practical realities of building, testing, and refining a full-fledged, real-world web application. Several important lessons emerged over the course of the project, spanning technical, design, and team-collaboration dimensions.

On the technical front, working with a MERN-style stack (React.js, Node.js, Express.js, and MongoDB) reinforced the importance of maintaining a clean separation between frontend and backend logic, particularly when building features involving multiple interdependent states, such as the job negotiation system, where offers, counter-offers, and negotiation history all had to remain perfectly synchronized between the client and freelancer views in real time. Implementing the escrow-based wallet mechanism, covering fund holding, milestone-based release, platform fee deduction, and withdrawal handling, highlighted how critical it is to plan data models carefully in advance, since a poorly structured schema for transactions or milestones can quickly lead to inconsistencies, such as the milestone-total validation issue encountered during testing.

The debugging phase in particular reinforced the value of rigorous, feature-by-feature testing rather than relying solely on surface-level checks. Issues such as the blank payment settings page, the mismatch between the finalized negotiated amount and the amount displayed in the project workspace, and the failed auto-redirection after project posting were the kind of bugs that were easy to overlook during initial development but became immediately apparent once each user flow was tested end-to-end from the perspective of an actual freelancer or client. This experience underscored those state management issues, particularly around values that can be updated after their initial creation, such as a negotiated project amount, require deliberate handling throughout the entire application rather than being assumed to propagate correctly by default.

From a design perspective, building out features such as the live chat system, contracts dashboard, and spending analytics view made clear how much real usability depends on presenting complex information, like transaction summaries, milestone progress, and negotiation history, in a simplified, digestible format for users who are not necessarily technically inclined. This pushed the team to prioritize UI clarity alongside backend functionality, rather than treating the interface as an afterthought.

Finally, working as a five-member team throughout the project reinforced the importance of clear module ownership, consistent coding conventions, and regular integration testing, since features developed independently, such as the negotiation system and the escrow wallet, ultimately needed to interact seamlessly within shared pages like the project workspace. Overall, this project provided hands-on experience in translating a set of academic objectives into a functioning, tested, and genuinely usable product, an experience that proved far more instructive than following a purely theoretical development process.

CHAPTER 8: RECOMMENDATIONS AND FUTURE SCOPE

8.1 Recommendations

Based on the development experience, testing outcomes, and limitations identified over the course of this project, several practical recommendations can be made for improving the platform's readiness, reliability, and adoption potential before any real-world or commercial deployment is considered.

Firstly, it is recommended that the platform undergo a dedicated round of load and stress testing prior to public release, simulating concurrent usage by a large number of freelancers and employers, particularly around high-traffic operations such as project search, bidding, and payment processing, to identify and resolve any performance bottlenecks that were not apparent during academic-scale testing. Secondly, given that payment security and compliance are especially sensitive in a platform handling real financial transactions, it is strongly recommended that direct UPI integration and automated tax documentation (such as TDS handling) be prioritized early in any post-submission development roadmap, ideally in consultation with a fintech partner or payment gateway provider experienced in Indian regulatory requirements.

It is also recommended that the platform introduce a formal dispute resolution mechanism, rather than relying solely on direct chat-based communication between freelancers and employers, to handle disagreements over deliverables, deadlines, or payment amounts in a structured and fair manner, potentially involving a designated support or moderation team. Additionally, before wider rollout, the AI-based recommendation engine should be validated and refined using a larger, more diverse dataset of real user interactions, rather than relying purely on the limited data generated during development and testing, to ensure recommendation accuracy improves meaningfully at scale.

From an accessibility standpoint, it is recommended that multi-language support be considered a near-term priority rather than a distant future enhancement, given the platform's specific focus on serving the Indian freelancing ecosystem, including users from Tier 2 and Tier 3 cities who may be more comfortable working in regional languages. Finally, it is recommended that platform-specific business rules, such as fee percentages and wallet balance expiry periods, be made configurable through an administrative panel rather than hardcoded, allowing the platform's operators to adjust these parameters over time based on real usage patterns and business needs, without requiring changes to the underlying codebase.

Collectively, these recommendations aim to bridge the gap between the platform's current state as a successfully tested academic project and its potential evolution into a fully deployable, production-ready system suitable for real-world use.

8.2 Future Scope

While the current implementation of the platform successfully fulfils its core objectives, there remains substantial scope for future enhancement across several dimensions, ranging from advanced AI capabilities to broader platform integrations and expanded market reach. The following areas have been identified as promising directions for future development.

Advanced AI Features

Building on the existing AI-based recommendation engine, future iterations of the platform could incorporate proactive skill-gap analysis, wherein the system identifies gaps between a freelancer's current skill set and the skills most in demand within their chosen category, offering personalized suggestions for upskilling or certification. Similarly, AI-assisted project scoping could be introduced to help employers who may not have technical expertise articulate clear, well-structured project requirements simply by describing their needs in plain language, with the system automatically generating a structured job posting. Predictive analytics could also be developed to estimate the likelihood of a project's successful completion based on historical data, freelancer track record, and project complexity, helping employers make more informed hiring decisions.

Resume and Profile Automation

A particularly valuable enhancement would be the introduction of resume PDF parsing, allowing freelancers to upload an existing resume or CV, from which the system would automatically extract and populate relevant profile fields, such as skills, work experience, education, and certifications, significantly reducing the time and effort required to set up a detailed profile manually. This could be extended further with automatic portfolio suggestions, where the system scans uploaded work samples or linked repositories (such as GitHub) to auto-generate portfolio entries, and periodic profile-completeness prompts that encourage freelancers to keep their profiles updated and competitive.

Localization and Payment Integration

To better serve the Indian freelancing ecosystem, future versions of the platform should prioritize direct UPI integration for instant, low-friction payments and withdrawals, alongside automated TDS compliance and tax documentation to simplify financial reporting for freelancers. Multi-language support, particularly for regional languages relevant to Tier 2 and Tier 3 cities, would further improve accessibility and adoption among a broader base of Indian freelancers who may not be fully comfortable operating in English.

Platform Expansion

Looking further ahead, the platform could expand into industry-specific vertical marketplaces, such as dedicated sections for legal consulting, academic tutoring, or creative arts, each tailored with category-specific profile fields, pricing norms, and search filters. The development of native mobile applications for iOS and Android would significantly improve accessibility for users who prefer managing freelance work on the go, while advanced collaboration tools, such as shared file workspaces, integrated video calling, and real-time collaborative document editing, could deepen the platform's utility beyond simple job matching and communication.

Additional Enhancements

Beyond the areas above, several other enhancements could further strengthen the platform over time. These include a formal dispute resolution and mediation system to handle disagreements over deliverables or payments in a structured manner, an admin panel allowing configurable business rules such as platform fees and wallet expiry periods, integration with third-party identity verification services to reduce fraud and build additional user trust, and API access for third-party integrations, allowing the platform to connect with external tools such as accounting software or project management systems commonly used by freelancers and small businesses.

Collectively, these future enhancements represent a clear roadmap for evolving the platform from a successfully tested academic project into a comprehensive, production-ready solution capable of competing meaningfully within India's growing freelance and gig economy.

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16. MongoDB Official Documentation - <https://www.mongodb.com/docs>
17. Tailwind CSS Official Documentation - <https://tailwindcss.com/docs>
18. Postman Learning Center - <https://learning.postman.com>

General References

19. W3Schools - Web Development Documentation (HTML, CSS, JavaScript) - <https://www.w3schools.com>
20. Stack Overflow - Developer Community Discussions and Troubleshooting - <https://stackoverflow.com>

APPENDICES / ANNEXURES

Annexure A - Project Development Timeline

The project was developed over a period of 38 weeks (November 2025 - July 2026), following a phased development approach. The work progressed from requirement gathering and literature review , through core infrastructure setup (MERN stack) , feature-by-feature development (authentication, job posting, escrow payments, real-time chat) , and concluded with comprehensive testing, deployment, and final documentation.

Phase-wise Summary

Phase	Duration	Weeks	Key Deliverables
1. Planning & Requirement Analysis	3 Nov - 26 Nov 2025	Week 1-4	Problem identification, literature review of existing platforms (Upwork, Fiverr) , user requirement gathering , and tech stack finalization (React, Node.js, MongoDB).
2. Project Setup & Architecture	22 Nov - 14 Dec 2025	Week 3-6	3-tier client-server architecture design , database schema planning , UI/UX wireframing for dashboards, and GitHub repository initialization.
3. Core UI & Authentication	10 Dec 2025 - 18 Jan 2026	Week 6-11	Landing page, client/freelancer role selection , registration/login UI, JWT authentication integration , and user profile management modules.
4. Backend Integration & Job Management	15 Jan - 8 Apr 2026	Week 11- 23	Express.js server setup , MongoDB Atlas connection , CRUD operations for job postings, proposal submission workflows, and negotiation interfaces.
5. Payment Gateway & Escrow Integration	2 Apr - 12 May 2026	Week 22-28	Razorpay/Stripe integration , secure escrow wallet logic , milestone verification tracking , and financial transaction summary dashboard.
6. System Testing & Bug Fixing	8 May - 16 Jun 2026	Week 27-33	API testing via Postman , layout alignment fixes , resolving the blank payment settings page , and enforcing milestone amount validation.

7. Deployment & Optimization	12 Jun - 4 Jul 2026	Week 32-36	Cloud hosting deployment (AWS/Render) , database query optimization, fixing auto-redirection post-job publishing, and final UI refinements.
8. Final Review & Documentation	1 Jul - 18 Jul 2026	Week 35-38	User Acceptance Testing (UAT) , drafting the final project report , preparing presentation slides, and final academic submission.

Annexure B: Development Tools, Technologies & Package Versions

To ensure a robust, scalable, and maintainable platform, the following tools, frameworks, and package versions were utilized during the development of MasterHire.

1. Frontend (Client-Side) Technologies

- **React.js (v18.2.0):** Used as the core JavaScript library for building the dynamic, component-based user interface.
- **Tailwind CSS (v3.4.1):** Utilized as the primary utility-first CSS framework for rapid UI styling and responsive design.
- **React Router DOM (v6.22.0):** Managed single-page application (SPA) navigation and protected route handling.
- **Axios (v1.6.7):** Handled asynchronous HTTP requests to the backend REST APIs.
- **Framer Motion / Anime.js:** Used for subtle UI animations and smooth page transitions.

2. Backend (Server-Side) Technologies

- **Node.js (v20.11.0 LTS):** Provided the asynchronous, event-driven JavaScript runtime environment for the server.
- **Express.js (v4.18.2):** Served as the web application framework for structuring the RESTful APIs and middleware logic.
- **JSON Web Token / JWT (v9.0.2):** Implemented for secure, stateless user authentication and session management.
- **Bcrypt.js (v5.1.1):** Handled cryptographic password hashing to securely store user credentials.

- **Socket.io (v4.7.4):** Powered the real-time, bi-directional communication for the platform's chat and notification systems.

3. Database & Storage

- **MongoDB (v7.0):** Acted as the primary NoSQL, document-oriented database.
- **Mongoose (v8.1.1):** Object Data Modelling (ODM) library used to enforce schema structures, validation, and complex querying.
- **MongoDB Atlas:** The fully managed cloud database service used for hosting the production database.
- **Multer / AWS S3 (or Cloudinary):** Handled the uploading, storage, and retrieval of freelancer portfolio files and project deliverables.

4. AI / Machine Learning & Analytics

- **Python (v3.12):** The core programming language used for the backend recommendation engine.
- **Scikit-learn (v1.4.0):** Utilized for skill-similarity scoring and collaborative filtering algorithms.
- **Pandas (v2.2.0):** Handled data manipulation and analytical processing for platform-wide metrics.

5. Third-Party APIs & Services

- **Razorpay Node SDK / Stripe API:** Integrated to securely process fiat transactions, escrow holdings, and user wallet withdrawals.
- **Google OAuth 2.0:** Integrated for frictionless, one-click sign-up and login capabilities.

6. Development, Testing & Design Tools

- **Visual Studio Code (v1.89):** The primary Integrated Development Environment (IDE) used by the team for writing and managing the codebase.
- **Git (v2.44.0):** The local version control system used for tracking code changes, branching, and maintaining project history.
- **GitHub (Cloud/SaaS):** The remote hosting platform used for collaborative code management, version syncing, and maintaining the central repository.

- **Postman (v10.24):** Utilized extensively for manual backend testing, API request formulation, and endpoint verification.
- **Figma (Desktop v116.17 / Web):** Used for initial UI/UX wireframing, prototyping, and layout planning before frontend development began.
- **Chrome DevTools (Chrome v124):** The built-in browser toolset used for frontend debugging, network request monitoring, and CSS layout inspection.

Annexure C: Core API Endpoints Summary

The MasterHire platform utilizes a RESTful API architecture built with Express.js and Node.js. This backend serves as the communication bridge between the React.js frontend, the MongoDB database, and external services (like Razorpay/Stripe and the Python AI engine).

HTTP Method	API Endpoint	Description	Auth Required
Authentication & User Management			
POST	/api/auth/register	Registers a new Client or Freelancer account.	No
POST	/api/auth/login	Authenticates a user and returns a JWT session token.	No
GET	/api/users/profile	Fetches the logged-in user's dashboard and profile data.	Yes
PUT	/api/users/profile/update	Updates user details, skills, and portfolio uploads.	Yes
Job Management			
POST	/api/jobs/create	Allows a Client to publish a new job posting.	Yes (Client)
GET	/api/jobs/search	Retrieves job listings based on applied search filters and categories.	Yes
GET	/api/jobs/:jobId	Retrieves the full description and details of a specific project.	Yes
Proposals & Negotiation			
POST	/api/proposals/submit	Submits a freelancer's initial bid and cover letter for a job.	Yes (Freelancer)

PUT	/api/proposals/:id/negotiate	Records a counter-offer in the negotiation history.	Yes
PUT	/api/proposals/:id/accept	Formalizes the agreement, changes status to 'Hired', and locks the price.	Yes (Client)
Escrow & Payment Handling			
POST	/api/payments/wallet/topup	Initiates a wallet deposit via Razorpay/Stripe gateway.	Yes (Client)
POST	/api/payments/escrow/deposit	Deducts from Client wallet and locks funds in escrow for a milestone.	Yes (Client)
POST	/api/payments/escrow/release	Releases escrow funds to the Freelancer minus platform fees.	Yes (Client)
POST	/api/payments/wallet/withdraw	Processes a Freelancer's request to transfer earnings to their UPI/Bank.	Yes (Freelancer)
Messaging & File Sharing			
GET	/api/messages/:projectId	Fetches the real-time chat history for an active project workspace.	Yes
POST	/api/files/upload	Uploads project deliverables securely to cloud storage (e.g., AWS S3).	Yes